

Gang Takedowns and Neighborhood Violence: Evidence from a Case Study in Philadelphia*

Hyeongeun Choe¹, Alyssa Mendlein², and Aaron Chalfin^{1,3}

¹Department of Criminology, University of Pennsylvania

²University of Pennsylvania Crime & Justice Policy Lab

³National Bureau of Economic Research

REVISED DRAFT

September 21, 2026

Abstract

Research Summary

Between 2018 and 2021, the Philadelphia District Attorney’s Office convened a series of special grand juries to investigate gang- and group-based violence within a single community encompassing four Census tracts in South Philadelphia. The investigations culminated in two coordinated “gang takedowns” and resulted in the arrest and prosecution of 21 individuals charged with serious offenses, 18 of whom were subsequently convicted and incarcerated. Although these individuals constituted just 0.1% of the community’s residential population, their removal was followed by a substantial decline in violence. Over the subsequent three years, serious violence in the focal area declined by between 20 and 50% relative to comparable Philadelphia communities with similar pre-intervention crime trends. Notably, the reduction in gun violence occurred without the displacement of crime to adjacent neighborhoods.

Policy Implications

These findings suggest that the targeted arrest and incapacitation of a very small number of high-risk individuals can produce large and sustained reductions in lethal violence. This paper highlights the value of targeted strategies that concentrate investigative and prosecutorial resources on individuals who occupy a central position in violence-producing networks.

Keywords: gang enforcement, synthetic controls

*We are indebted to individuals at the Gun Violence Task Force for the access to data and detailed explanations of the work that we are evaluating. We thank John MacDonald for helpful comments. Correspondence: Aaron Chalfin, achalfin@sas.upenn.edu.

1 Introduction

Violent crime, particularly lethal violence, is heavily concentrated in a small number of locations (Weisburd, 2015). Within these places, violence is further clustered among a relatively small group of predominantly young men who are embedded within tightly connected social networks in their communities (Braga & Clarke, 2014; Papachristos et al., 2012, 2013). These individuals often occupy positions within informal gangs or groups that structure patterns of conflict, retaliation, and alliance and which are therefore responsible for a disproportionate share of serious violence in many disadvantaged urban neighborhoods (Barao et al., 2021; Chalfin et al., 2021). The dynamics raise an important question: can law enforcement meaningfully reduce serious violence by identifying a small number of the most central actors within these social networks and targeting them for surgical enforcement actions?

In principle, there is much to recommend such a strategy. If violence is, in fact, driven by what Sherman (2007) has termed the “power few,” then interventions that incapacitate these actors – or deter others on the periphery – may generate outsized public safety benefits. Removing key individuals from active networks may also disrupt retaliatory cycles, alter expectations about the consequences of violence and reduce opportunities for future conflict. If successful, such an approach holds the promise of reducing serious violence without substantially expanding the reach of the criminal justice system. That is, it may be possible to achieve meaningful reductions in violence by arresting a small – or possibly very small – number of the right individuals rather than by relying on the use of the criminal justice system broadly.

These ideas align with a broader shift in policing and crime prevention toward strategies that concentrate enforcement resources on the most predictable risks. One prominent branch of this literature examines hot spots policing, which uses crime data and spatial analysis to identify micro-places with disproportionately high levels of crime and allocate police resources accordingly. A large body of experimental and quasi-experimental evidence indicates that focusing police presence on these high-risk locations can produce meaningful reductions in crime without merely displacing offending to nearby areas (Braga & Bond, 2008; Jabri, 2021; Sherman & Weisburd, 1995). The success of hot spots policing underscores an important principle: crime is not randomly distributed, and interventions that target highly concentrated risks can generate outsized, albeit still fairly modest, returns (Braga et al., 2019).

Complementing place-based strategies, a growing body of research emphasizes the importance of targeting high-risk individuals and groups within crime hot spots. In many cities, violence is not only spatially concentrated; it is also socially structured, emerging from networks of individuals linked through friendship, kinship, neighborhood ties and gang affiliation. Network-based theories of violence emphasize that conflict

often spreads through processes of retaliation and contagion within these social structures (Papachristos et al., 2012). Individuals connected to victims or offenders face elevated risks of subsequent victimization or offending and violence can therefore propagate through these networks over time. From this perspective, interventions that focus on key actors within violent networks may have large and possibly at-scale effects if they interrupt cycles of retaliation or alter behavioral expectations within the network.

Gang- and group-focused strategies leverage the existence of these networks and represent a particularly salient example of focused, precision policing. Such strategies have a long history in the United States, dating back at least a century (Hureau et al., 2023), though contemporary interventions vary widely in both their design and underlying philosophy. Some interventions emphasize preventive and supportive services that are intended to address the social and economic risk factors associated with gang involvement and include educational supports, cognitive behavioral therapy and employment initiatives, among other types of services (Bhatt et al., 2024). These interventions aim to reduce violence by steering high-risk youth away from criminal activity and toward more stable life trajectories. Other approaches rely more heavily on enforcement tools, including civil gang injunctions (Ridgeway et al., 2019) and large-scale gang sweeps in which law enforcement simultaneously arrests multiple members of a gang or criminal group (Chalfin et al., 2021; Domínguez, 2026; Ratcliffe et al., 2017). Between these differing approaches lies the strategy of focused deterrence, which combines targeted enforcement with offers of social services and explicit communication about the legal consequences of continued violence (Braga & Kennedy, 2013; Braga et al., 2018; Tillyer & Kennedy, 2008).

Despite their focus on different underlying mechanisms, these strategies share a common premise: that the risk of involvement in serious violence, either as a victim or perpetrator, is highly concentrated among a small and often predictable set of individuals (Heller et al., 2024). If these individuals can be accurately identified, interventions that concentrate enforcement or prevention resources on them may produce substantial reductions in violence. While there is a mature literature on hot spots policing (Braga et al., 2019) and focused deterrence (Braga et al., 2026), there is less causal evidence on the extent to which removing a very small number of highly active individuals from a community can generate sustained reductions in lethal violence.

In this paper, we study a series of highly targeted enforcement actions undertaken by law enforcement officials in Philadelphia, the sixth-largest city in the United States. Between 2018 and 2021, the Philadelphia District Attorney’s Office convened a series of special investigative grand juries to examine gang- and group-based violence within a single community encompassing four Census tracts in South Philadelphia. These investigations, undertaken in partnership with the Philadelphia Police Department and the

Pennsylvania Attorney General’s office, culminated in two coordinated “gang takedowns” that resulted in the arrest and prosecution of 21 individuals charged with serious offenses, 18 of whom were subsequently convicted and incarcerated. We study the effect of the two takedowns by comparing crime in the focal area in which the targeted gangs were based, with crime in other Philadelphia neighborhoods that experienced similar public safety trends prior to the takedowns but which were not the target of focused gang enforcement. To do so, we rely on two related methods: the synthetic differences-in-differences estimator of Arkhangelsky et al. (2021) and the synthetic controls estimator of Abadie et al. (2010). Both methods provide a data-driven way to identify an appropriate comparison group and each method provides a means of generating p -values using placebo-based inference.

Recognizing that the intervention’s focal area – where law enforcement believed that most of the crimes committed by the focal groups have taken place – is likely incomplete, we utilize unique administrative data that are infrequently available to researchers: location data extracted from cell phones owned by defendants who were the subject of the investigations. Using these data, we are able to observe where defendants were spending time prior to their arrests and generate a more sophisticated test of crime spillovers where we take cell phone locations, last known addresses and incident locations into account. We complement this analysis with traditional tests of displacement using areas adjacent to the focal takedown area.

Although the 21 arrested individuals constituted just 0.1% of the community’s roughly 20,000 residents, their removal was followed by a substantial decline in violence. Over the subsequent three years, serious violence in the focal area declined by between 20% and 50% relative to comparable Philadelphia communities with similar pre-intervention crime trends. Notably, the reduction in gun violence occurred without detectable displacement of offending to adjacent neighborhoods. These findings suggest that the targeted arrest and incapacitation of a very small number of high-risk individuals can produce large and sustained reductions in lethal violence. Importantly, there is less evidence that property crimes or less serious violence were affected, underscoring the considerable concentration of serious violence among criminal networks in contrast to property offending which is more broad-based.

The remainder of the paper proceeds as follows. Section 2 describes the special investigative grand juries and the subsequent gang takedowns. Section 3 details the administrative datasets that we assembled for the analysis. Section 4 outlines the empirical strategy. Section 5 presents the results, and Section 6 concludes.

2 Institutional Background

Philadelphia has implemented several innovative strategies to address gun violence including efforts emerging from the city’s recent expansion of the Gun Violence Task Force (GVTF). The GVTF is a multi-agency partnership comprising prosecutors and analysts from the Philadelphia District Attorney’s Office (DAO), investigators from the Philadelphia Police Department (PPD) and attorneys from the Pennsylvania Office of Attorney General. One particularly novel strategy pursued by the GVTF has been the use of the state’s investigating grand jury system to develop conspiracy cases against individuals who are believed to be involved in group-based retaliatory violence.

Grand juries have been utilized in the United States since colonial times, initially to assess probable cause in criminal prosecutions and investigate government wrongdoing (Ouziel, 2022). In modern times, grand juries are used to provide indictments for serious criminal charges, which are required in about half of the states and the federal system and optional in others. Grand juries consist of between 16 and 23 jurors who are tasked with determining whether there is enough evidence presented by the prosecutor to believe there is probable cause that an individual committed a crime, unlike petit (trial) juries that determine guilt or innocence.

In Pennsylvania, grand juries are authorized under the Investigating Grand Jury Act (Pa.C.S. §§4541-4553) and are limited to terms of no more than 24 months. Unlike traditional charging grand juries, investigating grand juries function as extended investigative bodies that assist prosecutors in developing complex criminal cases while also protecting citizens from unfounded criminal charges. They have the authority to compel testimony and the production of evidence through subpoenas and operate under strict secrecy requirements designed to preserve the integrity of ongoing investigations. These grand juries are often used in complex, long-term cases involving organized crime, public corruption, police misconduct and other conspiratorial offenses in which evidence may be spread across multiple actors and incidents.

Although the GVTF had previously used the grand jury process, it had not been applied to prosecute group-related violence in Philadelphia. The approach had, however, been used by Pennsylvania prosecutors in other complex cases – including a large-scale investigation into sexual abuse by members of the clergy. GVTF prosecutors consulted with attorneys involved in that case to better understand how the grand jury process could be adapted to cases involving long-running retaliatory violence with historical roots. Several features made the approach particularly attractive including the ability to compel testimony under conditions of secrecy and to aggregate evidence across multiple incidents into a unified conspiracy framework.

Two of the earliest cases using this strategy – which form the basis of this research – involved feuding groups in South Philadelphia. Beginning in 2018, a GVTF prosecutor worked with detectives from the PPD South Division to identify unsolved shootings believed to be connected to retaliatory group violence. Their review identified a sharp increase in shootings associated with individuals linked to several neighborhood groups. Further investigation suggested that these smaller groups had consolidated into two larger factions: 1) the 27th Street group (including members of 27th & Dickinson, the Wilson Park group, CTM, and 13th & Saigon) and 2) the 31st Street group (including 31st & Tasker, 18th Street, and HBlock).

Although the neighborhood had a history of rivalries, testimony suggested that the most recent conflict could be traced to an interpersonal dispute in 2015 that subsequently escalated into a broader factional divide. Investigators concluded that the conflict was driven less by drug markets than by status competition and retaliatory dynamics, often fueled by social media taunting and the circulation of videos referencing prior shootings. Ultimately, investigators identified at least 30 shootings connected to the feud, often occurring in alternating cycles across each group’s territories. The persistence and interconnection of these incidents motivated the GVTF to pursue a grand jury strategy focused on conspiracy and group liability rather than treating the shootings as isolated events.

Using more conventional investigative approaches, the prosecutor initially attempted to synthesize existing evidence across incidents, identifying links among individuals, vehicles and shooting locations. However, this evidence was initially considered insufficient to support criminal charges. Progress accelerated following a car theft involving four individuals previously identified as group members. A community policing officer helped locate the suspects who were subsequently stopped while in possession of the stolen vehicle — an incident which allowed investigators to recover and ultimately gain access to their cell phones. Around the same time, two additional shootings occurred. Collaboration between an experienced detective and the GVTF prosecutor helped refine the investigative strategy, including the development of targeted search warrants. These efforts served to establish the contours of a conspiracy involving multiple group members, although the identity of specific shooters remained unclear at that stage.

Despite these limitations, prosecutors determined that there was sufficient evidence to initiate a grand jury investigation. One early procedural lesson was the importance of staggering subpoenas. Initially multiple witnesses were subpoenaed simultaneously which alerted potential targets to the investigation. Over time, however, investigators assembled a substantial body of evidence, including digital evidence recovered from cell phones (such as text messages and social media content), call detail records (CDRs) used to approximate suspect locations, ballistic evidence linking firearms to more than 30 shootings, forensic evidence from stolen vehicles, and witness testimony describing the group conflict. Digital evidence has proven espe-

cially important, including communications in which individuals discussed their involvement in shootings and connections among co-conspirators. Surveillance footage also helped link suspects to incidents through firearms, distinctive clothing, and vehicles, which could be cross-referenced with photographs and videos recovered from phones and social media accounts.

The investigating grand jury ultimately focused on nine incidents with the strongest overlapping evidence and recommended charges in two cases, both from December 2017. The first involved the fatal shooting of a 27th Street member inside a Chinese takeout restaurant by two individuals later identified as members of the 31st Street group. The second involved the shooting of two 31st Street members who had been lured to a residence through communications from members posing as an unaffiliated female. Phone records from the “luring phone” connected the incident to two 27th Street members, and investigators concluded the attack was retaliatory. In total, six 27th Street members were determined to be responsible, including the two shooters.

In February 2019, the grand jury recommended charges against eight individuals, including charges for murder, criminal conspiracy, attempted murder, aggravated and simple assault, violations of the Uniform Firearms Act (VUFA), recklessly endangering another person, possession of an instrument of crime and criminal use of a communication facility. Additional charges were filed against several individuals for conduct that impeded the investigation, including hindering apprehension, obstruction of justice, evidence tampering and one perjury charge. Arrests occurred in March and April 2019, with all but one defendant detained pretrial. Seven ultimately entered guilty pleas (five open pleas and two negotiated pleas), with sentences imposed largely in 2021. The time from arrest until plea bargain ranged from approximately nine months to three and a half years.

Information obtained during these arrests generated a second phase of the investigation. Individuals agreed to cooperate with law enforcement and their testimony, corroborated through independent evidence, supported a second grand jury investigation. This grand jury reviewed 18 additional shootings, most targeting 27th Street members or associates. These cooperating witnesses provided critical information linking specific individuals to firearms and identifying shooters, while also helping prosecutors contextualize the broader group conflict.

This testimony was corroborated through multiple forms of evidence, including CDR analysis, digital evidence from seized phones, social media communications and videos, NIBIN ballistic matches linking firearms across incidents, surveillance video (including footage from the city’s Real Time Crime Center camera network) and forensic evidence such as DNA and fingerprints recovered from stolen vehicles. The grand jury also reviewed evidence of group membership, relying on six categories of indicators: (1) self-

identification, (2) social media promotion of the group, (3) participation in music videos advocating group identity or violence, (4) repeated involvement as victims or witnesses in group-related shootings, (5) ballistic linkages, and (6) forensic cell phone analysis.

This second grand jury investigation concluded in April 2021 with recommended charges against 15 individuals (including two already in custody from the first takedown) for their roles in 18 shootings occurring between August 2016 and October 2020. Charges included murder, conspiracy to commit murder, attempted murder, aggravated and simple assault, reckless endangerment, terroristic threats, firearms violations, possession of an instrument of crime, receiving stolen property and unauthorized use of a motor vehicle. For each incident, all identified participants were charged according to their roles, often resulting in multiple charges per defendant and counts multiplied by the number of victims.

Arrests began in April 2021. Eleven defendants were detained prior to their trial dates, while two spent at least some time released pending trial. Among the 13 newly charged individuals, eight entered guilty pleas (six negotiated and two open pleas), two were convicted at trial, and most were sentenced around 2023, with case processing times ranging from approximately four months to two and a half years. Of the remaining defendants, one case was adopted federally, one was withdrawn, and one remains pending as of March 2026 due to overlapping prosecutions in other jurisdictions.

Table 1 summarizes case outcomes for individuals charged in each of the two grand jury investigations. Across the two discrete takedowns, 21 individuals were arrested and 18 were eventually convicted of at least one criminal offense. All of those convicted served time in jail or prison, with sentences that ranged from 11.5-24 months to 90 years. In total, those convicted during the takedowns were sentenced to serve between 219 and 434 years in prison, an average of between 12 and 24 years per convicted defendant.

3 Data

Data on the activities of the GVTF – as well as geolocation information and court outcomes for defendants – were obtained through the use of an embedded researcher. These data include the boundaries of the focal area associated with the gangs targeted by the intervention, incident locations, last known residential addresses of defendants arrested during the takedowns, and call detail records (CDRs) that provide additional information on where defendants spent time prior to their arrests. These data are essential for studying the effects of the gang takedowns because, given the potential mobility of targeted individuals, there is some uncertainty regarding the geographic areas in which any resulting crime reductions should

be expected.¹ Each data source is described below.

Gang territory boundaries were provided by the lead prosecutor based on investigative knowledge of the feuding gangs’ activities and the locations of incidents examined by the special grand jury. One gang was targeted during the first takedown and the rival gang during the second. Their territories are geographically proximate and together span four Census tracts in South Philadelphia.

We supplement the prosecutor’s expertise with several additional data sources. First, we use the addresses of the 21 criminal incidents (20 shootings and 1 gun charge) for which offenders were charged in the takedowns to document the locations where gang members were alleged to have committed serious crimes. These locations were compiled from grand jury presentments listing recommended charges.² Second, last known addresses for each offender charged in the takedowns were obtained from the state court data management system. Because this system records addresses at the time of entry, it often contains multiple addresses for the same individual across time. We retain all available addresses for the 22 individuals in our sample, yielding a total of 98 unique addresses used in the analysis.³

Third, we incorporate call detail records (CDRs), which document communication events generated when mobile phones initiate or receive calls, send text messages, or access the internet. These records include anonymized identifiers for the caller and receiver, the connected cell tower IDs for both parties, call duration, and timestamps (Okmi et al., 2023). CDRs are commonly used in criminal investigations to corroborate a suspect’s location near the time of an incident and to establish general mobility patterns surrounding alleged offenses.⁴ The CDRs used in this study were introduced as evidence in the prosecuted cases and were subsequently compiled and aggregated by a GVTF analyst to identify the most frequent locations of phone activity for individuals involved in the takedowns. Although CDRs are an imperfect proxy for mobility due to their event-driven nature and heterogeneity in phone usage patterns (Zhao et al., 2016), they provide valuable supplementary information about where the intervention’s impacts may plausibly occur following the removal of these individuals from the street. GVTF provided CDR summaries for 11 individuals, including the ten most frequently used cell towers for each individual based on call volume. While these data are not available for all group members, we treat them as a proxy for identifying areas where gang members were most active.

¹If offenders are strongly tied to specific places, as suggested by prior research (Weisburd et al., 2006), any treatment effects would likely be highly geographically concentrated. However, if offenders are relatively mobile, it becomes important to identify the broader set of locations where they spent time prior to the intervention.

²Incident location data are limited for the first takedown (two incidents), while 18 incidents are observed for the second takedown between 2016 and 2020.

³One individual included in the last known data had died before the investigative grand jury charged the individual.

⁴CDRs have also been used in criminological research to predict crime, identify mobility patterns, and detect criminal networks (Okmi et al., 2023).

We combine CDRs, last known addresses and incident locations to construct an aggregate measure of the areas in which targeted individuals spent time prior to their arrests. This composite measure is used to identify potential spillover areas and produces a broader geography of gang activity that extends beyond the four focal Census tracts identified by the District Attorney’s Office. To create such a measure, the city of Philadelphia was divided into thousands of small cells (216 x 234 meters), and a kernel density function was used to create the intensity measure for gang activity.⁵ The weights were applied to CDRs and last known address data.⁶ The standardized densities from each source are summed up and aggregated at the Census tract level, a process which is summarized in Panel (a) of Figure 1. In the main analysis, we exclude the top 30 Census tracts ranked according to the composite gang activity and their adjacent tracts to account for potential spillover effects and ensure an uncontaminated comparison group.

Figure 1 presents a map of Philadelphia showing the focal intervention area and additional locations where targeted offenders were active. Panel (a) presents the aggregate measure of expected intervention exposure, while Panels (b)–(d) display the individual components: incident locations (Panel b), call detail record locations (Panel c), and last known addresses (Panel d). The incident and CDR data indicate that gang activity was highly concentrated within a small geographic area of South Philadelphia spanning four Census tracts. However, both the CDRs and residential address data suggest that a broader set of Census tracts—primarily also located in South and Southwest Philadelphia—may plausibly experience spillover effects. These patterns motivate our examination of treatment effects both within and beyond the focal intervention area.

We supplement the takedown and court outcome data with two additional sources. First, we obtained crime incident data from 2015 through 2024 from OpenDataPhilly.⁷ Crime incidents are aggregated to the Census-tract-by-quarter level. We examine several (partially overlapping) crime categories: (1) Violent crime, defined as the sum of homicides, rapes, aggravated assaults, robberies, and shootings; (2) Gun-involved violence, defined as aggravated assaults or robberies involving a firearm; and (3) Property crime, defined as the sum of burglary, theft, motor vehicle theft, and arson. In auxiliary analyses, we also examine specific crime categories separately, including shooting victims, motor vehicle theft, and other assaults.

Finally, we incorporate neighborhood-level covariates from the American Community Survey (ACS) five-year estimates for each year from 2015 to 2024. Census tract-level variables include population size,

⁵The kernel density surface at each location is estimated using the Gaussian kernel with bandwidth 500 meters, $\hat{f}(x, y) = \frac{1}{2\pi * 500^2} \sum_{i=1}^n \exp\left(-\frac{(x-x_i)^2 + (y-y_i)^2}{2 * 500^2}\right)$.

⁶For the CDRs, locations with a higher rank with respect to call volume receive greater weight in the analysis. For the last known address, where individuals frequently had multiple recorded addresses, weights were defined as the inverse of the total number of addresses associated with each individual.

⁷<https://opendataphilly.org/datasets/crime-incidents/>

racial composition, gender and age distributions, unemployment rate, median household income, poverty rate, foreign-born population share, and the share of households receiving public assistance. Because Census tract boundaries were updated following the 2020 Census, we use data from the National Historical Geographic Information System (NHGIS) to harmonize tract definitions over time and ensure geographic consistency throughout the study period.

4 Methods

This paper examines the effect of two gang takedowns within a community in South Philadelphia that encompasses four Census tracts. For the purposes of evaluation, we consider these four Census tracts to be a single treatment area that was subject to the intervention while many other communities in Philadelphia remained predominantly unexposed to the intervention. This yields an opportunity to study the gang takedowns using a comparative case study framework — an extension of the traditional approach of differences-in-differences to studying place-based crime control policies. In the context of a case study with a single treated unit, the norm in the literature is to use one of two approaches: the canonical synthetic controls estimator of Abadie et al. (2010) or the newer synthetic differences-in-differences estimator of Arkhangelsky et al. (2021). In this section we describe both frameworks and how we implement them to study the effects of the Philadelphia gang takedowns. As a robustness check, we also estimate traditional differences-in-differences models using ordinary least squares regression.

In the context of an intervention that occurs at a single point in time and which affects multiple exposed units, the standard DiD estimator is implemented using a simple linear regression model:

$$Y_{it} = \alpha + \beta_1 TREAT_i + \beta_2 POST_t + \beta_3 (TREAT_i \times POST_t) + \varepsilon_{it} \quad (1)$$

In (1), β_3 represents the differences-in-differences estimate of the intervention’s impact which, in the absence of covariates, is equal to the average change in the outcome variable among the treated units minus the average change in the outcome variable among the comparison units. The estimate is valid under the assumption of parallel trends which is partially testable using an event study. There are several main challenges to using the canonical DiD estimator to evaluate a case study in which there is only a single treated unit — or a small number of treated units. First, statistical inference becomes problematic because standard errors derived from large sample asymptotics rely on having many treated units. With only a single treated unit, conventional clustering approaches may understate uncertainty. Second, the credibility of the parallel trends assumption is more difficult to assess since the counterfactual trend for the treated

unit must be inferred from a potentially heterogeneous set of control units whose pre-treatment dynamics may not closely match those of the treated case. Third, and relatedly, the estimated treatment effect may be highly sensitive to the particular choice of a comparison group, raising concerns about the robustness of the resulting estimates.

4.1 Synthetic Controls for Case Studies

The method of synthetic controls for case studies, introduced by Abadie et al. (2010), has substantially changed how researchers approach quantitative case studies, particularly in settings involving a single treated unit and a large set of possible control units. This framework is especially relevant for policy evaluations at the city or regional level, an important context in criminal justice research in which a central challenge is determining an appropriate comparison group.

Historically, researchers often justified comparison units based on geographic similarity or a resemblance in baseline characteristics. In practice, this meant selecting a comparison group based on theoretical considerations and then relying on the hope that the parallel trends assumption, the key identifying condition underlying differences-in-differences designs, would approximately hold. Synthetic control methods improve on this approach by constructing a *synthetic* comparison unit that closely reproduces the treated unit’s pre-intervention trajectory — thereby replacing ad hoc selection with a more transparent and structured procedure and one in which the parallel trends assumption is more likely to be met. The approach now features prominently in empirical work in the social sciences. Examples in criminology include studies of immigration and crime (Chalfin & Deza, 2020), police turnover (Mourtgos et al., 2022), police use of force (Goh, 2021), crime intelligence centers (Lawrence et al., 2026), death penalty moratoriums (Oliphant, 2022), labor market shifts (Mitre-Becerril & Chalfin, 2021), place-based interventions (Buggs et al., 2022; Lawrence, 2023; Lawrence et al., 2022; Piza et al., 2020; Robbins et al., 2017; Rydberg et al., 2018; Saunders et al., 2015), prosecutorial reforms (Hogan, 2022; S. Wu & McDowall, 2024; Zhou et al., 2023), marijuana liberalization (Harper & Jorgensen, 2023; G. Wu & Cullenbine, 2024), and gun control policies (Donohue et al., 2019).

The method works by constructing a comparison group using a weighted average of units that are unexposed to the intervention, with weights chosen to minimize the difference in pre-intervention trends between the treated unit and its “synthetic comparison.” Consider a panel dataset with one treated unit (indexed by 0) and N untreated donor units observed over T periods, with T_0 pre-intervention periods. Let $Y_{it}(1)$ and $Y_{it}(0)$ denote the potential outcomes with and without treatment. Since only unit 0 is treated after T_0 , we observe $Y_{0t} = Y_{0t}(0)$ for $t \leq T_0$ and $Y_{0t} = Y_{0t}(1)$ for $t > T_0$, while $Y_{nt} = Y_{nt}(0)$ for all donor

units $n \geq 1$. The treatment effect in period T is:

$$\tau_T = Y_{0T}(1) - Y_{0T}(0), \quad (2)$$

where the counterfactual $Y_{0T}(0)$ is unobserved. The synthetic control method (SCM) estimates this counterfactual by constructing a weighted average of donor units with weights $\mathbf{W} = (w_1, \dots, w_N)$ chosen to match the treated unit's pre-intervention characteristics. SCM is typically motivated by a linear factor model for untreated outcomes (Abadie et al., 2010):

$$Y_{it}(0) = \delta_t + \boldsymbol{\theta}_t^\top \mathbf{Z}_i + \boldsymbol{\lambda}_t^\top \boldsymbol{\mu}_i + \epsilon_{it}, \quad (3)$$

where δ_t are common time effects, $\mathbf{Z}_i$ are observed covariates, $\boldsymbol{\mu}_i$ are unobserved factors, and ϵ_{it} are mean-zero shocks. If weights exist such that the synthetic control reproduces the treated unit's observed and unobserved determinants of outcomes, SCM can approximate the missing counterfactual.

Given weights $\mathbf{W}$, the counterfactual is estimated as:

$$\hat{Y}_{0T}(0) = \sum_{n=1}^N w_n Y_{nT}, \quad (4)$$

which yields the estimator:

$$\hat{\tau}_T = Y_{0T} - \sum_{n=1}^N w_n Y_{nT}. \quad (5)$$

Weights are then selected to minimize the distance between the treated and synthetic covariates:

$$\left(\sum_{k=1}^K v_k \left(X_{0k} - \sum_{n=1}^N w_n X_{nk} \right)^2 \right)^{1/2}, \quad (6)$$

subject to $w_n \geq 0$ and $\sum_{n=1}^N w_n = 1$. These constraints improve interpretability and prevent extrapolation — though bias may still arise if predictors are weakly related to outcomes or if the factor structure is misspecified (Kellogg et al., 2021). Subtracting the average pretreatment differences from the average post-treatment differences yields the DiD estimates for SCM specifications. Statistical significance is determined by a placebo inference procedure⁸.

⁸The placebo inference method proposed by Abadie et al. (2010) is analogous to a permutation test. We apply the SCM method to each unit in the donor pool as if it had received the intervention and obtain a corresponding DiD estimate. This procedure allows us to assess whether the estimated treatment effect for the target tracts is large relative to the distribution of placebo effects obtained by assigning the intervention to untreated units. The p -value is calculated based on the rank of

4.2 Synthetic Differences-in-Differences

While SCM provides a data-driven approach for constructing a counterfactual comparison unit, obtaining a reliable synthetic control may be difficult when the treated unit’s outcome levels differ substantially from those of many donor units. To address this limitation, Arkhangelsky et al. (2021) proposes the synthetic difference-in-differences (SDiD) estimator, which combines features of traditional differences-in-differences designs with the weighting structure of SCM.

The SDiD approach offers several potential advantages relative to standard SCM. First, it relaxes the need for a very close pre-treatment level match by incorporating unit fixed effects, which allow for persistent level differences between the treated and control units during the pre-intervention period. This feature is particularly useful in settings where no convex combination of donor units can closely reproduce the treated unit’s outcome levels. It is also more consistent with the identifying assumption of differences-in-differences which requires parallel trends but does not require equivalent baseline levels. Second, SDiD introduces time weights that allow the estimator to place greater emphasis on pre-treatment periods that are most informative for predicting post-treatment outcomes. By optimally reweighting time periods, the estimator can reduce bias and improve efficiency by limiting the influence of pre-treatment periods that are less comparable to the post-treatment environment (Arkhangelsky et al., 2021). Finally, SDiD incorporates regularization when estimating the unit and time weights, which helps prevent overfitting, particularly in settings with a large donor pool.

The SDiD estimator can be written as the solution to the following minimization problem:

$$(\hat{\tau}^{\text{sdid}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau, \mu, \alpha, \beta} \sum_{i=1}^N \sum_{t=1}^T \hat{\omega}_i^{\text{sdid}} \hat{\lambda}_t^{\text{sdid}} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2. \quad (7)$$

This objective function resembles a weighted two-way fixed effects regression, where α_i and β_t denote unit and time fixed effects, respectively, and $\hat{\omega}_i^{\text{sdid}}$ and $\hat{\lambda}_t^{\text{sdid}}$ are the estimated unit and time weights. The inclusion of unit fixed effects allows the estimator to account for baseline level differences between treated and control units, while the time weights emphasize the most informative pre-treatment periods for constructing the counterfactual trend. The regularization used in constructing these weights further helps stabilize estimation and reduce sensitivity to noise. Given these advantages—including improved pre-treatment fit in settings with level differences and potentially more stable estimation—we use SDiD as our primary specification. We also report SCM estimates as a robustness check.

the estimated effect for the treated tracts relative to the placebo estimates, divided by the total number of sample units.

4.3 The Present Study

The present study compares the evolution of crime in the four Census tracts that comprise the focal area of the intervention to other areas in Philadelphia experiencing similar pre-intervention crime trends. In order to generate estimates, we consider the intervention to have begun in March 2019 when the first of the two takedowns occurred. Since the second takedown occurred in the aftermath of the first takedown, we are unable to formally disambiguate between the effects of the first takedown and the second takedown. Instead, we visually present the evolution of crime trends throughout the post study period and provide treatment effects separately for the period just prior to the second takedown and for the entire study period.

For each outcome and quarter, we define the outcome as the mean across the four Census tracts targeted by the gang takedowns. We construct the donor pool by excluding areas that are unsuitable for comparison. First, we exclude the tracts adjacent to the focal area to account for potential spatial crime displacement. Second, because a separate special investigative grand jury targeted a community in North Philadelphia during the post-intervention period, we exclude the six tracts targeted by subsequent gang takedowns and their nine adjacent tracts. Third, using cell phone mobility data that track the movements of defendants charged in the intervention, we exclude the 30 Census tracts in which defendants spent the most time, along with their adjacent tracts. Finally, to ensure baseline comparability and avoid comparisons with areas experiencing little violent crime, we restrict the donor pool to tracts with quarterly violent crime counts above the citywide median. Figure 2 illustrates these restrictions using a map of Philadelphia. The four focal Census tracts are shown in dark blue. The 30 tracts with the highest composite activity levels among defendants, together with their adjacent tracts, are shown in light blue. These areas are excluded from the donor pool and analyzed separately under an auxiliary treatment definition. The six tracts targeted by subsequent gang takedowns and their nine adjacent tracts (purple) are also excluded. In practice, we further restrict the donor pool by excluding Census tracts with violent crime rates below the citywide mean (the gray areas). The remaining tracts are eligible for inclusion in the donor pool, subject to the violent-crime restriction described above. The final eligible comparison tracts are shown in red.

One additional restriction is worth noting. Following advice in Abadie et al. (2010), we ultimately eliminate the placebo comparisons whose pretreatment match is not successful by excluding the placebo units whose root mean square prediction error (RMSPE) is greater than that of twice the treatment unit. In a series of robustness checks, we re-analyze our data and test the sensitivity of our estimates to each of these restrictions.

Finally, we directly examine the spillover effects of the takedowns. The spillover effects can be observed in both directions given that the takedowns might have either displaced crime into spillover areas or spread the positive effects across potential spillover areas. Our conceptualization of spillover areas includes the adjacent tracts to the target area and the top 6 active tracts (See Figure 3) in terms of the constructed metric that we compiled from three different sources.⁹

5 Results

5.1 Main results

We begin by presenting descriptive evidence on the effect of the gang takedowns. Figure 4 plots the percentage change in violent crime between 2015 and 2024 across Philadelphia’s 384 Census tracts. The percentage change is computed by comparing the average number of violent crimes between the pre period (2015- 2018) and the post period (2019 - 2024). Over the study period, roughly half of the tracts experienced increases in violent crime, while the remaining half experienced declines. A smaller subset of tracts — approximately one-eighth — exhibited large increases in violence, whereas only a very small share experienced comparably large reductions. The four tracts highlighted in red correspond to those targeted by the gang takedowns. These tracts are ranked 5th, 27th, 31st, and 40th in the distribution of tract-level changes in violence, indicating that each falls roughly within the upper decile of tracts experiencing the largest reductions in violence during the period.

We construct a heuristic inferential benchmark to assess the likelihood of observing such a pattern by chance. Specifically, drawing four tracts without replacement from a donor pool of 384 tracts yields $\binom{384}{4} = 891,881,376$ possible combinations. We then ask how likely it is that four randomly selected tracts would exhibit ranks at least as extreme as those observed in the data. Summing the observed ranks — 5, 27, 31, and 40 — yields a total rank of 103. The probability that four randomly drawn tracts would have a rank sum of 103 or lower is approximately 0.018 percent or about 1 in 5,600.¹⁰ Taken together, these results suggest that the treated tracts constitute an unusually strong concentration of crime reductions relative to what would be expected under random assignment across the city during the period of the gang takedowns.

Next, we present formal estimates from synthetic differences-in-differences models, which, due to its

⁹We initially intended to include the top 10 tracts in terms of the metric, but four of them are found to be the target areas, yielding the 6 remaining tracts.

¹⁰Though a closed form solution to this problem is feasible using combinatorics, we sidestep that analytic complexity and generate a very close estimate of this probability via simulation, using 1,000,000 iterations of four random draws from a distribution of 384 ranked objects.

greater analytic flexibility and precision compared to SCM, is our preferred estimator. The sample that we begin with includes 15,360 observations ($384 \text{ census tract} \times 40 \text{ quarters}$) between 2015-2024. Table 2 presents the average quarterly crime count and neighborhood information for Philadelphia, treatment tracts, and comparison tracts during the study period. There were 14.7 police-reported violent crime incidents per quarter in the Census tracts that were exposed to the intervention, compared to 9.8 in Philadelphia as a whole. As mentioned earlier, we restrict our comparison tracts to those Philadelphia Census tracts in which the level of violent crime is above the median of Philadelphia such that the comparison pool excludes the areas with low-baseline violent crime. As a result of this, the donor pool that we use to generate SDiD and SCM estimate is more comparable to the four target tracts. The donor pool, on average, experiences 15.7 violent crimes per tract-quarter. Notably, the target area and the donor pool are quite similar demographically with nearly identical median incomes and similar rates of poverty and receipt of public assistance.

Table 3 reports the result of our primary SDiD specifications of the effect of the gang takedowns for different crime types. Each outcome variable is transformed via the inverse hyperbolic sine transformation, an approximation of the natural logarithm which accommodates zero values (Norton, 2022) and so the table presents the estimated treatment effects, standard errors and 95% confidence intervals in percentage terms. For context, we report these estimates alongside the pre-intervention mean for the focal tracts. Our findings indicate that the gang takedowns led to a qualitatively large and statistically significant reduction in overall violent crimes (-48%), gun-involved violence (-43%) and shooting victims (-34%). Reflecting the relative rarity of crimes within a small area, the confidence intervals are wide. However, the estimates are robust to a wide array of different specifications, which we establish in the next section of the paper. Auto theft is examined as one of the outcomes based on the expectation that gangs (and specifically the two involved in the takedowns) commit a large share of motor vehicle theft. While the point estimate for auto theft was negative, the estimates are not sufficiently precise to rule out a null effect. Consistent with expectations, no significant change is observed in property crimes and other assaults, crimes which are committed by a much larger number of individuals.

For each outcome variable, Figure 5 presents the estimates visually, providing a means of assessing the quality of the pre-intervention fit for the SDiD estimator. In each of the panels, the vertical dashed lines indicate the quarter at which each of the two takedowns occurred. The shaded area in the bottom of the panels represents the time weights assigned by the SDiD estimator. The sloped dashed line represents the counterfactual trend for the treated tracts in the absence of the takedowns. Pre-treatment fit is assessed

using the root mean squared prediction error (RMSPE).¹¹

5.2 Diffusion of Benefits and Displacement

Next, we consider whether the intervention may have caused crime to change in other areas of Philadelphia. There are two primary ways in which this could have happened. First, since defendants, in fact, spent time outside of the focal area, there is a question of whether the crime reductions observed in the focal area may have also occurred in other areas where defendants spent time prior to being arrested. If so, estimated treatment effects for the focal area will tend to understate the benefits of the intervention. Second, there is the standard concern that police activity in the focal area could have caused crimes to be displaced to outlying areas. While the majority of the literature does not find evidence of displacement effects when it comes to police-led place-based interventions (Braga et al., 2019), this is a prospect that nevertheless requires empirical engagement. In light of each of these possibilities, we identify two potential sets of areas in which to test for diffusion of benefits and displacement: (a) the top 6 tracts in which defendants spent the greatest amount of time based on last known addresses and call detail records and (b) the 8 Philadelphia Census tracts that were adjacent to the target tracts.

Table 4 presents the results of both analyses. The top panel (Panel a) suggests that the intervention led to a 29% decrease both in violent crime and gun-involved violence and a 35% decrease in shooting victims in six Census tracts outside of the focal area in which the defendants were most active. The analysis provides evidence that the crime-reducing benefits of the intervention are not limited to the target area but also spill over to the areas where defendants were spending an appreciable amount of time. Interestingly, the decline in auto theft in these areas was also significant, a result that may indicate that defendants stole vehicles outside of the focal area to use in crimes committed within the focal area. As an analog to Figure 5, Figure A1 and Figure A2 present the estimates visually. The bottom panel presents the analysis of the adjacent tracts to the target tracts. Each of the six estimates is negative and two — shootings victims and auto thefts — are statistically significant. The estimates suggest that shooting victims and auto theft decreased by 22% and 28%, respectively.

¹¹For each pre-treatment period, we calculated the difference between the observed outcome for the treated unit and the outcome of the synthetic comparison constructed using the estimator’s unit weights. Because SDID allows for a constant level difference between the treated and synthetic units, we first removed the average pre-treatment gap and then calculated the square root of the mean squared residual difference across all pre-treatment periods. The raw RMSPEs range from 0.121 to 0.346. When compared to the corresponding average pre-treatment outcome levels in the treated tracts, most of the RMSPEs are small – as small as 3%. The RSMPE for shooting victims (27% of the average of the pre-treatment outcome) is considerably larger indicating a poorer fit. Here we note that the RMSPE decreases substantially to 9% when outcomes are aggregated at six-month intervals, suggesting that the poorer fit in the quarterly analysis is partly attributable to the sparsity of the outcome. Importantly, the estimated treatment effect remains substantively similar-and is, in fact, larger in magnitude than our main analysis-indicating a 50% reduction in shooting victims.

Together, these estimates suggest that two things are true. First, consistent with some of the prior literature that has studied place-based interventions, there is some evidence of diffusion of benefits to outlying areas. Second, there is very little evidence that crime has been pushed outward to these areas. Overall, the data suggest that crime reductions, while largest in the focal area, are not limited to that area and accrue both nearby and in other parts of Philadelphia where defendants are active.

Finally, we run an auxiliary and purely exploratory analysis in which we assess a particular hypothesis that came up in conversations with officials in the Philadelphia District Attorney’s Office — that it is suspected that members of the two focal gangs were in the habit of stealing vehicles from parking lots at Philadelphia International Airport that might then be used to commit other crimes. In Table A1 we report SDiD and SCM estimates for auto thefts at the Airport as a function of the timing of the initial gang takedown. Both estimates point to a potentially large decline in auto thefts at the airport after the intervention began, though estimates are too noisy to generate a high degree of confidence. The SDiD analysis estimates a 38% reduction in auto thefts at the airport ($p < 0.07$). The SCM estimate points to a 35% reduction in auto thefts at the airport ($p = 0.11$).

5.3 Robustness

Next, we consider the robustness of our findings to choices made during the process of preparing and analyzing the data. While robustness checks are always needed, we expect this to be especially true here as there are only a handful of treated units and sparse outcomes, both of which mean that the resulting estimates have relatively wide confidence intervals. The width of the confidence intervals leaves open the possibility that estimates might differ in important ways in different models. Research by Jelveh and Chalfin (2026) notably highlights the various ways in which case studies may be sensitive to analytic choices made during the research process. A related limitation in case study research is that there is always the possibility that the effects of the intervention can be confounded by other contemporaneous shocks. The purpose of this section is to address as many of these concerns as possible.

We begin by adding a richer set of covariates to the models, estimates that are presented in Table A2. This table presents SDiD estimates in which we additionally control for Census tract-level demographics including population, poverty rate, median household income, the unemployment rate, the share of households receiving public assistance and the share of the population that is White, male and foreign-born. Next, we add a vector of additional crime controls, meaning that we control for each crime type and not only the crime type on the left-hand side of the model as is the norm when estimating SDiD models. For instance, in the violent crimes model, we control for pre-intervention values of not only violent crimes but

also the other crime types. This results in a richer model that is used to achieve a pre-intervention fit. In each case, the estimates remain very close to our preferred estimates with estimates for violent crimes, gun-involved crimes and shooting victims retaining statistical significance.

Next, we repeat the same robustness checks for the spillovers analysis. Table A3 shows the robustness models for the top 6 Census tracts outside of the focal areas in which defendants were spending time. Consistent with the findings in Table 4, Panel A suggests the estimates for violent crime, gun-involved violence and shooting victims were consistent with the primary spillover analyses, although gun-involved violence was not quite statistically significant at $\alpha = 0.05$. Panel B in the table presents estimates for the adjacent tracts. The observed patterns in the main spillover analyses mostly hold for the robustness model, although the estimate for shooting victims was marginally significant. Table A4 and Table A5 present SCM estimates for the spillovers analyses. Estimates are noisier than the SDiD estimates but the pattern in the estimates remains very similar to that of our preferred estimates.

Next we consider robustness to the choice of estimator and report estimates using traditional differences-in-differences models using ordinary least squares regression (Table A6) and synthetic controls estimation (Table A7). The traditional differences-in-differences estimates are very close to those presented in Table 3, with estimates for violent crimes, gun-involved crimes and shooting victims retaining statistical significance. The synthetic controls estimates are likewise similar to our preferred estimates although they are estimated with slightly more noise. The estimate for violent crimes remains significant at conventional levels, whereas the estimates for gun violence (-42%) and shooting victims (-39%) remain large but are not quite significant at conventional levels — both p -values are 0.08. To assess pre-treatment fit, we provide the gap plots in Figure A3 for each outcome reported in Panel A of Table A7. The pre-treatment fit varies between the results, with RMSPE values ranging from 0.155 to 0.473. For their relative magnitude, RMSPEs range approximately from 4% to 37% of the corresponding average pre-treatment outcome levels in the treated tracts. The synthetic control for violent crime closely approximates the treated unit during the pre-treatment period relatively, whereas other sparser outcomes exhibit non-trivial deviation from the treated unit. Acknowledging that these deviations warrant caution in interpreting the SCM estimates, we present the SCM analyses as robustness checks rather than as our primary estimates.

We also report a host of additional robustness checks. In Table A8, we show how estimates vary when we engage in more aggressive trimming of the donor pool based on root mean square prediction error multiples of that of the treated unit, a recommendation that is made in Abadie et al. (2010). Estimates are similar regardless of how aggressively the sample is trimmed. In Table A9 and Table A10, we show how estimates vary when we change the inclusion criterion for the donor pool. In Table A9, we limit the donor

pool to the top 25% and the top 75% of tracts ranked by the number of violent crimes they experience as well as including all Census tracts in Philadelphia in the donor pool. Estimates are extraordinarily similar for each of the different inclusion thresholds. In Table A10, we alter the exclusion criteria to exclude even fewer or more Census tracts from the donor pool that might be touched by the intervention. In Panel A, we exclude the top 10 Census tracts and their adjacent tracts based on our measure of time spent in those tracts by defendants. In Panel B, we exclude the 50 Census tracts and their adjacent tracts. In Panel C, we exclude Kensington from the donor pool since Kensington, a major regional hub for drug sale and trafficking, has received a great deal of police attention during the study period. In all cases, estimates are very similar.

Finally, following recent place-based crime control studies (Chalfin et al., 2023; MacDonald et al., 2026; Mitre-Becerril, 2026), we aggregate the data into uniform hexagonal spatial units with 500-meter widths covering the entire city. This spatial reaggregation serves as a complement to our primary tract-level analysis by imposing a consistent areal unit and by addressing concerns regarding comparability across irregularly shaped and variably sized Census tracts. The resulting dataset consists of 1,854 hexagons, including 8 treatment hexagons and 703 comparison hexagons. Figure A4 displays the spatial delineation of the treatment and spillover areas using the hexagons. Panel (a) maps the 42 most active hexagons, which corresponds to an analogue of the “top 6 active tracts” specification reported in Table 4.¹² Panel (b) presents the 14 adjacent hexagonal areas, highlighting the spatial contiguity between the active and neighboring zones.¹³ Table A11 presents SDiD estimates using the hexagon-level specification. Panel A shows that the treatment-area estimates closely align with the baseline tract-level results. Panel B reports spillover effects for the top 42 active hexagons; these findings largely replicate those from the top 6 active tract specification in Table 4, with the exception that the reduction in shootings is no longer statistically significant. Panel C presents estimates for adjacent hexagon areas, where statistically significant reductions additionally emerge for violent crime and gun-involved violence, complementing the main spillover findings for shooting victims and auto theft.

¹²The choice of 42 hexagons is based on scaling the tract-level selection procedure by the relative number of spatial units. In the tract-level spillover analysis, the 10 most active tracts are first identified, excluding 4 treatment tracts to yield 6 comparison tracts. Given that the hexagon-level dataset ($N = 1,854$) contains approximately five times as many units as the tract-level dataset, we select the top 50 hexagons (10×5), exclude the 8 treatment hexagons, and obtain 42 active hexagons for comparability.

¹³Notably, the set of top 42 active hexagons includes adjacent tract areas.

6 Discussion

Between 2018 and 2021, the Philadelphia District Attorney’s Office convened a series of special grand juries to investigate gang- and group-based violence within a geographically concentrated community spanning four Census tracts in South Philadelphia. These investigations culminated in two coordinated gang take-downs that resulted in the arrest and prosecution of 21 individuals charged with serious violent and firearm offenses, 18 of whom were subsequently convicted and sentenced to terms of incarceration. Although these individuals represented approximately 0.1% of the community’s residential population, their removal was followed by a substantial and sustained decline in violence. Over the subsequent three years, serious violence — violent index crimes, shootings and gun-involved violence — in the focal area declined by between 20 and 50% relative to comparable Philadelphia communities exhibiting similar pre-intervention crime trends. Importantly, crime did not move around the corner: we find no evidence that violence was displaced to adjacent neighborhoods. If anything, there is evidence that benefits accrued to both adjacent areas as well as other areas of Philadelphia in which defendants had been spending time. Taken together, these findings suggest that the targeted arrest and incapacitation of a very small number of high-risk individuals can generate large and durable reductions in serious violence without producing obvious spatial displacement.

While this study focuses on a single case study, and thus any generalizations must be made with appropriate caution, the results are consistent with a growing body of research on more surgical or precision-focused gang enforcement. A number of recent evaluations similarly suggest that large public safety gains may be achievable through interventions directed at a very small number of chronically offending individuals who occupy central positions in violence-producing social networks. For example, recent research finds that a focused deterrence intervention in Baltimore’s Western Police District resulted in the arrest of approximately 0.2% of the district’s population and coincided with a roughly 30% reduction in shootings (Kapustin et al., 2026). Similarly, research by Chalfin et al. (2021) finds that the incapacitation of approximately 0.1% of the population in several high-crime public housing communities in New York City was associated with a roughly one-third reduction in shootings. In Los Angeles, *Operation Thumbs Down* resulted in 23 federal indictments and another 18 state arrests and led to a 22% reduction in violent crimes LAPD’s Southwest Division (Ratcliffe et al., 2017). Taken together, these findings reinforce a central empirical regularity in criminology: serious violence is highly concentrated among a small number of individuals and within a small number of co-offending networks. To the extent that this concentration can be reliably identified, it seems as though it is possible to leverage this empirical regularity to produce outsized improvements in public safety through highly targeted interventions.

At the same time, several important questions remain unresolved. First, while our findings are potentially consistent with an incapacitation-based explanation – in which the removal of 18 criminally active individuals directly reduced opportunities for violence – the extent to which the observed improvements reflect incapacitation as opposed to deterrence, network disruption or other types of peer effects remains uncertain. For example, enforcement actions of this sort may generate important deterrence effects that extend beyond those directly targeted, particularly among individuals on the periphery of criminally active groups who may update their perceptions about the risks of offending following highly visible prosecutions. Likewise, the takedowns may have affected the behavior of other groups operating in the same social or geographic space, particularly if those groups had ongoing conflicts with the targeted network. In such cases, enforcement actions could reduce cycles of retaliatory violence either by removing key actors or by temporarily suppressing inter-group conflict through heightened perceptions of legal risk. More generally, these interventions may work not only by removing individuals, but also by disrupting the relational structures through which violence is organized and transmitted.

Second, questions remain regarding both the scalability and the long-term sustainability of gang take-downs as a violence-reduction strategy. On the scalability margin, such interventions are resource-intensive and depend heavily on investigative capacity, interagency coordination and prosecutorial expertise. In Philadelphia, for example, the success of these investigations has relied, in part, on the District Attorney’s Office’s technical capacity to extract and analyze digital evidence from seized mobile devices pursuant to judicial warrants, as well as the ability to synthesize this information into prosecutable conspiracy cases. The DAO’s ability to extract data from phones has been made possible by discrete investments in these capabilities and through training attorneys and investigators on how to utilize this sort of evidence. Jurisdictions lacking comparable investigative resources or legal infrastructure may find it difficult to replicate these efforts at scale. However, part of this intervention is an adjustment to the role that prosecutors play in investigations. The GVTF has found that early career ADAs have been very amenable to this new role and excelled through training and supervision, and so the financial and technological barriers would be lower for other jurisdictions to adopt this aspect of the model.

Even if such interventions can be implemented elsewhere, questions remain about whether their effects can be maintained over longer time horizons. One concern, emphasized by Sharkey (2018), is whether enforcement-driven reductions in violence can produce a stable equilibrium in the absence of complementary social investments that address the structural conditions under which violence emerges. From this perspective, law enforcement may be capable of producing what Sharkey terms an “uneasy peace” – one that may prove fragile if not reinforced by investments in community organizations, employment oppor-

tunities and other stabilizing institutions. A related concern involves the long-term trajectories of the individuals who are incapacitated. Prior research raises the possibility that incarceration may, in some cases, reinforce criminal identities, strengthen gang affiliations, or deepen exposure to criminogenic networks, potentially increasing the risk of recidivism following release (Pyrooz et al., 2011). Whether targeted enforcement ultimately reduces lifetime offending among those directly affected therefore remains an open and important question. Finally, while this intervention was surgical and focused on a small number of individuals, there is always the risk of mission creep and an expansion of law enforcement presence in any area that is inconsistent with the original mandate.

These considerations point to the potential value of pairing surgical enforcement strategies with complementary interventions designed to promote desistance and strengthen community stability. Such approaches might include reentry programming (Jonson & Cullen, 2015), focused deterrence call-ins that combine enforcement with services (Braga et al., 2018; Roman et al., 2019), cognitive behavioral interventions (Bhatt et al., 2024), place-based approaches to violence reduction including greening vacant lots (Cui et al., 2022; Macdonald et al., 2022), remediating blighted properties (Branas et al., 2018; MacDonald, 2015), improved street lighting (Chalfin et al., 2022; Welsh et al., 2022) and, potentially, investments in community-based violence prevention organizations (Hureau & Papachristos, 2025; Solomon et al., 2025). From a policy perspective, the results presented here suggest that targeted enforcement can play a central important role in violence reduction but also highlight the need to situate such strategies as one component of a broader violence-reduction portfolio rather than as a standalone solution.

Finally, this study highlights the importance of continuing to build the evidence base around precision-focused enforcement strategies. Future research would benefit from additional multi-site evaluations, longer follow-up periods, and designs capable of disentangling incapacitation, deterrence, and network disruption mechanisms. Greater attention to heterogeneity in effects—both across communities and across types of violence-producing networks—would also help clarify the conditions under which such strategies are most likely to succeed. As jurisdictions continue to search for effective and politically viable approaches to reducing serious violence, careful empirical work on highly targeted interventions remains an important priority.

References

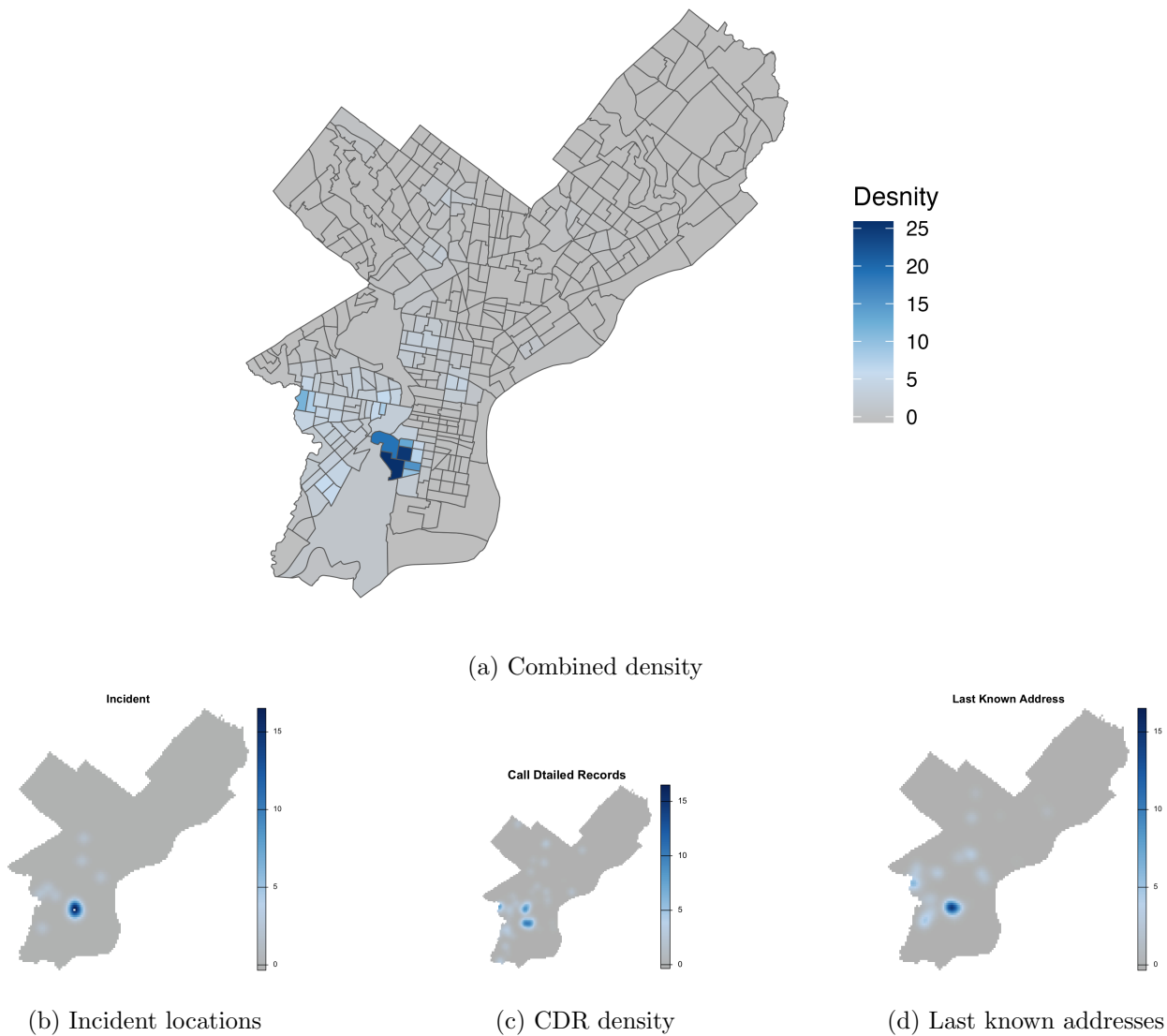
- Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of california’s tobacco control program. *Journal of the American Statistical Association*, 105(490), 493–505.
- Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic difference-in-differences. *American economic review*, 111(12), 4088–4118.
- Barao, L., Braga, A. A., Turchan, B., & Cook, P. J. (2021). Clearing gang-and drug-involved nonfatal shootings. *Policing: An International Journal*, 44(4), 577–590.
- Bhatt, M. P., Heller, S. B., Kapustin, M., Bertrand, M., & Blattman, C. (2024). Predicting and preventing gun violence: An experimental evaluation of readi chicago. *The Quarterly Journal of Economics*, 139(1), 1–56.
- Braga, A. A., & Bond, B. J. (2008). Policing crime and disorder hot spots: A randomized controlled trial. *Criminology*, 46(3), 577–607.
- Braga, A. A., & Clarke, R. V. (2014). Explaining high-risk concentrations of crime in the city: Social disorganization, crime opportunities, and important next steps. *Journal of Research in Crime and Delinquency*, 51(4), 480–498.
- Braga, A. A., & Kennedy, D. M. (2013). Linking situational crime prevention and focused deterrence strategies. In *The reasoning criminologist* (pp. 65–79). Routledge.
- Braga, A. A., Turchan, B. S., Papachristos, A. V., & Hureau, D. M. (2019). Hot spots policing and crime reduction: An update of an ongoing systematic review and meta-analysis. *Journal of Experimental Criminology*, 15(3), 289–311.
- Braga, A. A., Turchan, B. S., & Weisburd, D. L. (2026). Focused deterrence can reduce crime: A systematic review of randomized controlled trials and quasi-experiments. *Criminology & Public Policy*.
- Braga, A. A., Weisburd, D., & Turchan, B. (2018). Focused deterrence strategies and crime control: An updated systematic review and meta-analysis of the empirical evidence. *Criminology & Public Policy*, 17(1), 205–250.
- Branas, C. C., South, E., Kondo, M. C., Hohl, B. C., Bourgois, P., Wiebe, D. J., & MacDonald, J. M. (2018). Citywide cluster randomized trial to restore blighted vacant land and its effects on violence, crime, and fear. *Proceedings of the National Academy of Sciences*, 115(12), 2946–2951.
- Buggs, S. A., Webster, D. W., & Crifasi, C. K. (2022). Using synthetic control methodology to estimate effects of a cure violence intervention in baltimore, maryland. *Injury Prevention*, 28(1), 61–67.
- Chalfin, A., Del Pozo, B., & Mitre-Becerril, D. (2023). Overdose prevention centers, crime, and disorder in new york city. *JAMA network open*, 6(11), e2342228–e2342228.
- Chalfin, A., & Deza, M. (2020). Immigration enforcement, crime, and demography: Evidence from the legal arizona workers act. *Criminology & Public Policy*, 19(2), 515–562.
- Chalfin, A., Hansen, B., Lerner, J., & Parker, L. (2022). Reducing crime through environmental design: Evidence from a randomized experiment of street lighting in new york city. *Journal of Quantitative Criminology*, 38(1), 127–157.
- Chalfin, A., LaForest, M., & Kaplan, J. (2021). Can precision policing reduce gun violence? evidence from “gang takedowns” in new york city. *Journal of Policy Analysis and Management*, 40(4), 1047–1082.
- Cui, J., Jensen, S. T., & MacDonald, J. (2022). The effects of vacant lot greening and the impact of land use and business presence on crime. *Environment and Planning B: Urban Analytics and City Science*, 49(3), 1147–1158.
- Domínguez, M. (2026). Sweeping up gangs: The effects of tough-on-crime policies from a network approach. *American Economic Journal: Economic Policy*, 18(1), 127–158.

- Donohue, J. J., Aneja, A., & Weber, K. D. (2019). Right-to-carry laws and violent crime: A comprehensive assessment using panel data and a state-level synthetic control analysis. *Journal of Empirical Legal Studies*, 16(2), 198–247.
- Goh, L. S. (2021). Did de-escalation successfully reduce serious use of force in camden county, new jersey? a synthetic control analysis of force outcomes. *Criminology & Public Policy*, 20(2), 207–241.
- Harper, A. J., & Jorgensen, C. (2023). Crime in a time of cannabis: Estimating the effect of legalizing marijuana on crime rates in colorado and washington using the synthetic control method. *Journal of Drug Issues*, 53(4), 552–580.
- Heller, S. B., Jakubowski, B., Jelveh, Z., & Kapustin, M. (2024). Machine learning can predict shooting victimization well enough to help prevent it. *Review of Economics and Statistics*, 1–45.
- Hogan, T. P. (2022). De-prosecution and death: A synthetic control analysis of the impact of de-prosecution on homicides. *Criminology & Public Policy*, 21(3), 489–534.
- Hureau, D. M., Braga, A. A., Lloyd, T., & Winship, C. (2023). Streetwork at the crossroads: An evaluation of a street gang outreach intervention and holistic appraisal of the research evidence. *Criminology*, 61(4), 758–794.
- Hureau, D. M., & Papachristos, A. V. (2025). Re-centering the community in violence intervention: Re-claiming legacies of street outreach in the provision of public safety. *Annual Review of Criminology*, 8(1), 431–458.
- Jabri, R. (2021). Algorithmic policing. Available at SSRN 4275083.
- Jelveh, Z., & Chalfin, A. (2026). *Perils and pitfalls in the use of synthetic control methods to study public safety interventions* (tech. rep.). Working paper.
- Jonson, C. L., & Cullen, F. T. (2015). Prisoner reentry programs. *Crime and Justice*, 44(1), 517–575.
- Kapustin, M., Chalfin, A., Biddle, J., Layana, C., Wade, B. A., Struhl, B., & Braga, A. A. (2026). *Reducing gun violence without exacerbating the harms of policing in baltimore’s western district* (tech. rep.). Working paper.
- Kellogg, M., Mogstad, M., Pouliot, G. A., & Torgovitsky, A. (2021). Combining matching and synthetic control to tradeoff biases from extrapolation and interpolation. *Journal of the American Statistical Association*, 116(536), 1804–1816.
- Lawrence, D. S. (2023). Patrolling the largest drug market on the eastern seaboard: A synthetic control analysis on the impact of a police bicycle unit. *Criminology & Public Policy*, 22(3), 517–541.
- Lawrence, D. S., Peterson, B. E., & March, M. (2026). The relationship between crime information centers and crime: A micro-synthetic evaluation of a district-level policing strategy. *Journal of Criminal Justice*, 102, 102572.
- Lawrence, D. S., Peterson, B. E., Robin, L., & Shukla, R. (2022). The impact of correctional cctv cameras on infractions and investigations: A synthetic control approach to evaluating surveillance system upgrades in a minnesota prison. *Criminal Justice Policy Review*, 33(8), 843–869.
- Macdonald, J., Nguyen, V., Jensen, S. T., & Branas, C. C. (2022). Reducing crime by remediating vacant lots: The moderating effect of nearby land uses. *Journal of Experimental Criminology*, 18(3), 639–664.
- MacDonald, J. M. (2015). Community design and crime: The impact of housing and the built environment. *Crime and Justice*, 44(1), 333–383.
- MacDonald, J. M., Chalfin, A., Moritz, M., Wade, B., Mendlein, A. K., Braga, A. A., & South, E. (2026). Can enhanced street lighting improve public safety at scale? *Criminology & Public Policy*, 25(1), 31–62.
- Mitre-Becerril, D. (2026). The long-term effects of structural discrimination on public safety: The 1930s redlining maps. *Crime & Delinquency*, 00111287261425622.
- Mitre-Becerril, D., & Chalfin, A. (2021). Testing public policy at the frontier: The effect of the \$15 minimum wage on public safety in seattle. *Criminology & Public Policy*, 20(2), 291–328.

- Mourtgos, S. M., Adams, I. T., & Nix, J. (2022). Elevated police turnover following the summer of george floyd protests: A synthetic control study. *Criminology & Public Policy*, 21(1), 9–33.
- Norton, E. C. (2022). The inverse hyperbolic sine transformation and retransformed marginal effects. *The Stata Journal*, 22(3), 702–712.
- Okmi, M., Por, L. Y., Ang, T. F., & Ku, C. S. (2023). Mobile phone data: A survey of techniques, features, and applications. *Sensors*, 23(2), 908.
- Oliphant, S. N. (2022). Estimating the effect of death penalty moratoriums on homicide rates using the synthetic control method. *Criminology & Public Policy*, 21(4), 915–944.
- Ouziel, L. M. (2022). Prosecutors as partisans. *Fordham Urb. LJ*, 50, 1093.
- Papachristos, A. V., Braga, A. A., & Hureau, D. M. (2012). Social networks and the risk of gunshot injury. *Journal of Urban Health*, 89(6), 992–1003.
- Papachristos, A. V., Hureau, D. M., & Braga, A. A. (2013). The corner and the crew: The influence of geography and social networks on gang violence. *American Sociological Review*, 78(3), 417–447.
- Piza, E. L., Wheeler, A. P., Connealy, N. T., & Feng, S. Q. (2020). Crime control effects of a police substation within a business improvement district: A quasi-experimental synthetic control evaluation. *Criminology & Public Policy*, 19(2), 653–684.
- Pyrooz, D., Decker, S., & Fleisher, M. (2011). From the street to the prison, from the prison to the street: Understanding and responding to prison gangs. *Journal of Aggression, Conflict and Peace Research*, 3(1), 12–24.
- Ratcliffe, J. H., Perenzin, A., & Sorg, E. T. (2017). Operation thumbs down: A quasi-experimental evaluation of an fbi gang takedown in south central los angeles. *Policing: An International Journal of Police Strategies & Management*, 40(2), 442–458.
- Ridgeway, G., Grogger, J., Moyer, R. A., & Macdonald, J. M. (2019). Effect of gang injunctions on crime: A study of los angeles from 1988–2014. *Journal of quantitative criminology*, 35(3), 517–541.
- Robbins, M. W., Saunders, J., & Kilmer, B. (2017). A framework for synthetic control methods with high-dimensional, micro-level data: Evaluating a neighborhood-specific crime intervention. *Journal of the American Statistical Association*, 112(517), 109–126.
- Roman, C. G., Link, N. W., Hyatt, J. M., Bhati, A., & Forney, M. (2019). Assessing the gang-level and community-level effects of the philadelphia focused deterrence strategy. *Journal of experimental criminology*, 15(4), 499–527.
- Rydberg, J., McGarrell, E. F., Norris, A., & Circo, G. (2018). A quasi-experimental synthetic control evaluation of a place-based police-directed patrol intervention on violent crime. *Journal of Experimental Criminology*, 14, 83–109.
- Saunders, J., Lundberg, R., Braga, A. A., Ridgeway, G., & Miles, J. (2015). A synthetic control approach to evaluating place-based crime interventions. *Journal of Quantitative Criminology*, 31, 413–434.
- Sharkey, P. (2018). *Uneasy peace: The great crime decline, the renewal of city life, and the next war on violence*. WW Norton & Company.
- Sherman, L. W. (2007). The power few: Experimental criminology and the reduction of harm: The 2006 joan mccord prize lecture. *Journal of Experimental Criminology*, 3(4), 299–321.
- Sherman, L. W., & Weisburd, D. (1995). General deterrent effects of police patrol in crime “hot spots”: A randomized, controlled trial. *Justice Quarterly*, 12(4), 625–648.
- Solomon, S., Roman, C. G., Davey-Rothwell, M., Abaya, R., Webster, D., & Frattoroli, S. (2025). Community-based efforts to reduce violence: A scoping review on the implementation of cure violence. *INQUIRY: The Journal of Health Care Organization, Provision, and Financing*, 62, 00469580251360956.
- Tillyer, M. S., & Kennedy, D. M. (2008). Locating focused deterrence approaches within a situational crime prevention framework. *Crime Prevention and Community Safety*, 10(2), 75–84.
- Weisburd, D. (2015). The law of crime concentration and the criminology of place. *Criminology*, 53(2), 133–157.

- Weisburd, D., Wyckoff, L. A., Ready, J., Eck, J. E., Hinkle, J. C., & Gajewski, F. (2006). Does crime just move around the corner? a controlled study of spatial displacement and diffusion of crime control benefits. *Criminology*, 44(3), 549–592.
- Welsh, B. C., Farrington, D. P., & Douglas, S. (2022). The impact and policy relevance of street lighting for crime prevention: A systematic review based on a half-century of evaluation research. *Criminology & Public Policy*, 21(3), 739–765.
- Wu, G., & Cullenbine, R. R. (2024). Recreational marijuana legalization and drug-related offenses in washington state: An interrupted time series analysis with a combination of synthetic controls. *Journal of Experimental Criminology*, 20, 395–420.
- Wu, S., & McDowall, D. (2024). Does bail reform increase crime in new york state: Evidence from interrupted time-series analyses and synthetic control methods. *Justice Quarterly*, 41(3), 371–399.
- Zhao, J., Freeman, B., & Li, M. (2016). Can mobile phone apps influence people’s health behavior change? an evidence review. *Journal of medical Internet research*, 18(11), e287.
- Zhou, A., Koo, A., Kallus, N., Ropac, R., Peterson, R., Koppel, S., & Bergin, T. (2023). Synthetic control analysis of the short-term impact of new york state’s bail elimination act on aggregate crime. *Statistics and Public Policy*, 11(1).

Tables and Figures



Notes: Panel (A) shows the combined spatial density computed using incident locations, call detail records (CDR), and last known addresses. Panels (B)–(D) present the kernel density surface maps for each data source separately.

Figure 1: Spatial Density of Incident Locations, CDR Records, and Last Known Addresses

Treatment map

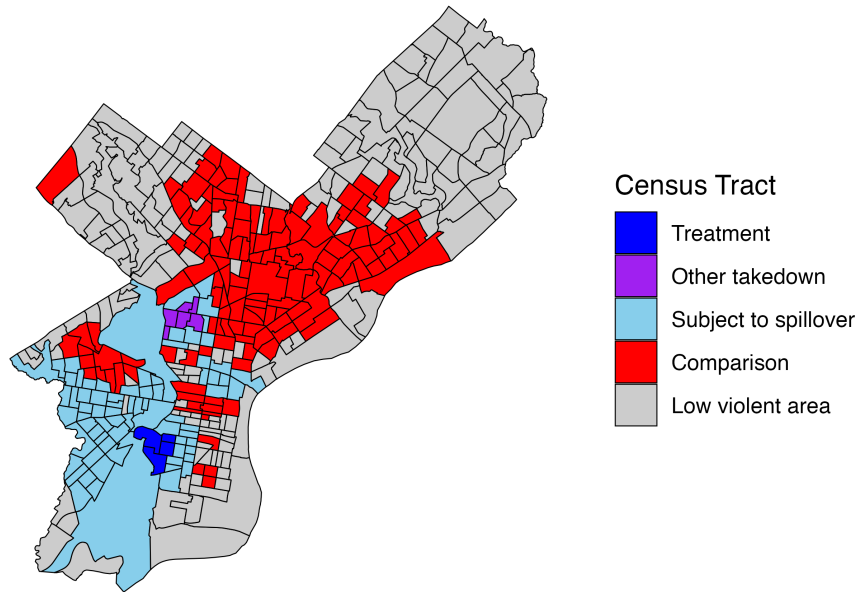


Figure 2: Map of treatment and comparison Census tracts

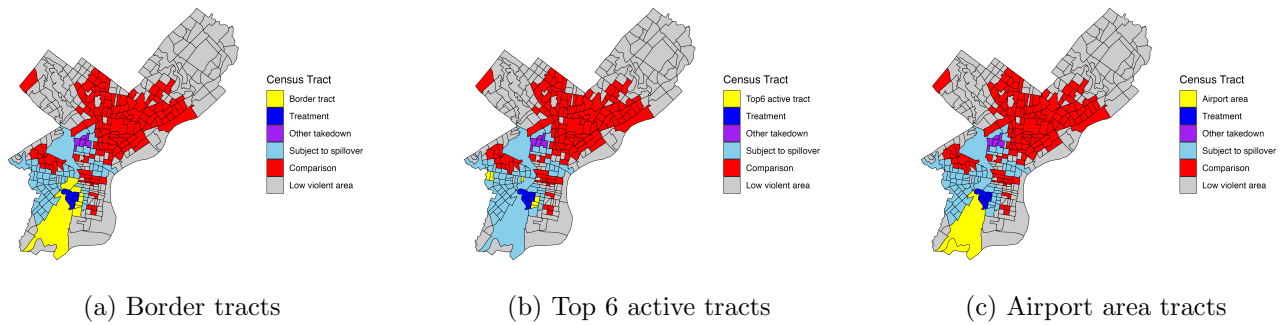
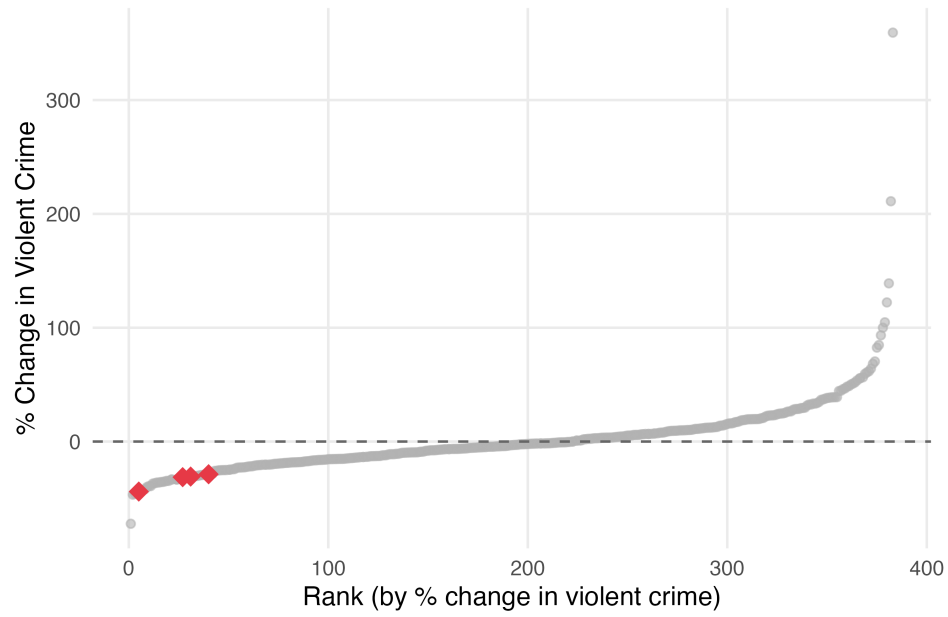
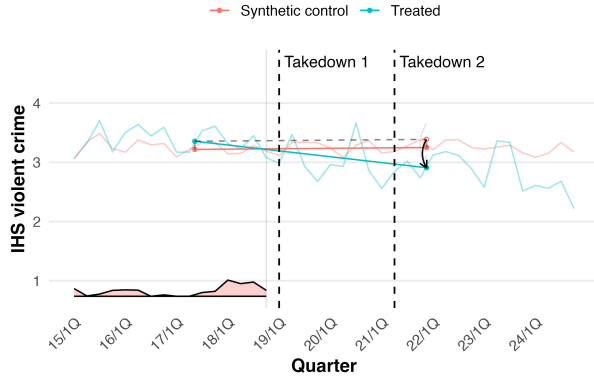


Figure 3: Spillover & Airport tract map

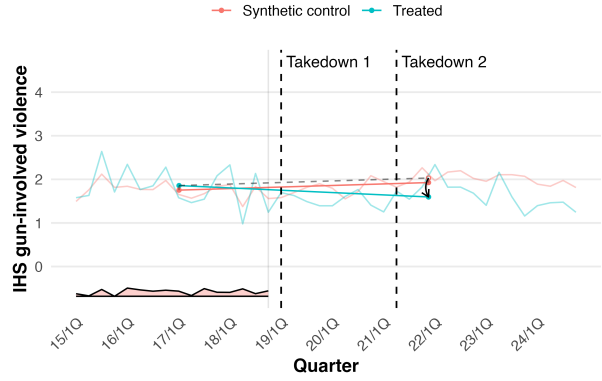


Notes: Lower % change indicates greater reduction. Red diamond represents the four treatment tracts. One outlying Census tract whose percent change in violent crime is exceptionally high is excluded from the plot.

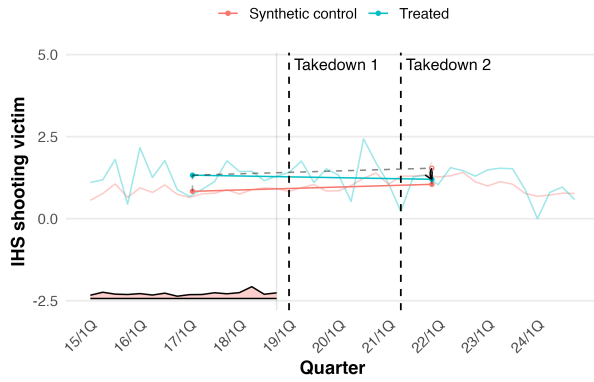
Figure 4: Distribution of Changes in Violent Crimes across tracts



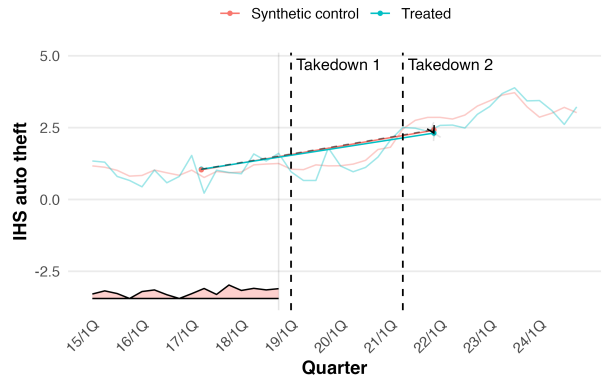
(a) Violent crime, RMSPE = 0.132



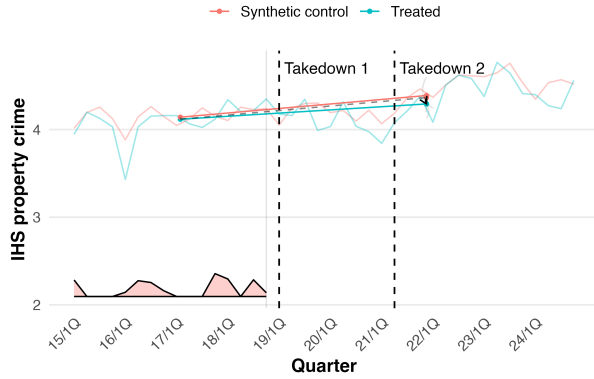
(b) Gun-involved violence, RMSPE = 0.275



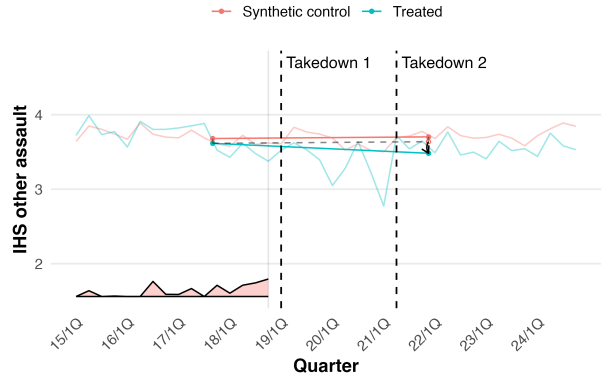
(c) Shooting victims, RMSPE = 0.346



(d) Auto theft, RMSPE = 0.280



(e) Property crime, RMSPE = 0.148



(f) Other assault, RMSPE = 0.121

Figure 5: Target Tracts Synthetic Difference-in-Differences Plots

Note. RMSPE denotes the pre-treatment root mean squared prediction error. Lower values indicate better pre-treatment fit.

Table 1: Grand Jury Outcomes

	Takedown 1	Takedown 2	Total
Census Population for Target Area			19,749 ^a
People Arrested	8	13 ^b	21
Number of Shooting Incidents Charged ^c	2	18	20
People Convicted	7	11	18 ^d
Sentence Range: Minimum	11.5 months – 30 years	11.5 months – 45 years	
Total Minimum	62.5 years	156.5 years	219 years
Sentence Range: Maximum	23 months – 60 years	23 months – 90 years	
Total Maximum	125 years	309.5 years	434.5 years
Avg. Number of Days Off the Street	1003.25	752.384	847.95 ^e

^a Sum of four census tracts during the pre-takedown period.

^b Two individuals were already in custody from Takedown 1 and received additional charges in Takedown 2; one individual is deceased.

^c The number of shooting incidents charged in investigative grand jury.

^d The one person who was not convicted in Takedown 1 had the charges nolle prossed. In Takedown 2, one case had charges withdrawn; another individual is still awaiting trial.

^e Conservative estimate, as some individuals were already incarcerated. This reflects additional incarceration time attributable to the takedowns as of 12/31/2024.

Table 2: Descriptive Statistics for Philadelphia, Target Tracts, and Comparison Tracts

Variable	Philadelphia	Target Tracts	Comparison Tracts
Number of tracts	384	4	133
Violent crime	9.79	14.70	15.70
Murder	0.04	0.04	0.06
Rape	0.59	0.85	0.90
Aggravated assault	3.52	3.71	5.82
Robbery	4.69	7.97	7.32
Gun-involved violence	2.72	3.64	4.40
Shooting	0.95	2.15	1.59
Shooting victims	1.06	2.36	1.77
Property crime	35.10	37.20	46.40
Auto theft	4.36	5.60	5.93
Population	4108	4884	4777
White	0.40	0.25	0.27
Unemployment	0.10	0.15	0.13
Male	0.47	0.47	0.47
Income	53650	46428	43276
Poverty	0.24	0.26	0.31
Foreign-born	0.13	0.09	0.13
Public assistance	0.07	0.08	0.09

Notes: This table reports the descriptive statistics for Philadelphia, target tracts, and comparison tracts. Crime variables are aggregated at quarterly level during the study period (2015 - 2024). Neighborhood variables are extracted from American Community Survey from 2015 to 2024. As Census tract boundaries are updated every 10 year and our tract definition is based on the 2010s, the ACS data since 2020 is incorporated using National Historical Geographic Information System.

Table 3: Synthetic Difference-in-Differences Estimates for Target Area

Outcome	Pre Mean	SDID (%)	SE (%)	95% CI (%)
Violent crime	18.12	-47.9	14.2	(-76, -20)
Gun-involved violence	4.33	-43.4	15.6	(-74, -13)
Shooting victims	2.42	-33.8	12.8	(-59, -9)
Auto theft	1.58	-11.7	14.2	(-40, 16)
Property crime	33.78	-7.3	13.9	(-35, 20)
Other assault	24.67	-15.5	12.1	(-39, 8)

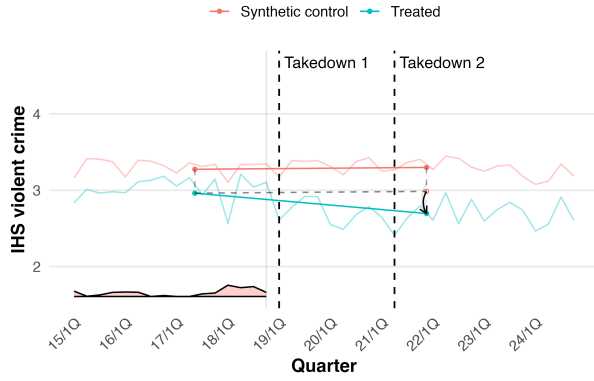
Notes: Pre Mean column denotes the quarterly crime on average prior to the intervention. Standard errors estimated using the placebo method.

Table 4: Synthetic Difference-in-Differences Estimates: Spillover Effects

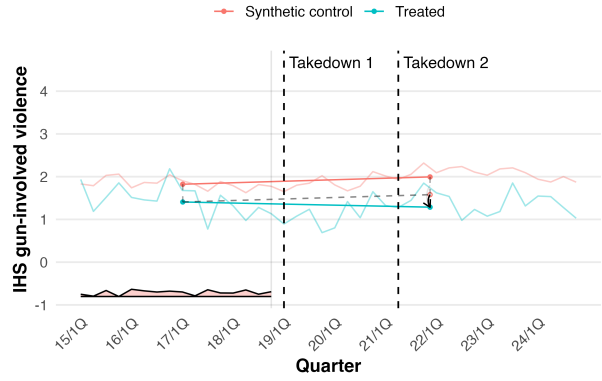
Outcome	Pre Mean	SDID (%)	SE (%)	95% CI (%)
Panel A: Top 6 Active Tracts				
Violent crime	13.85	-28.9	11.7	(-52, -6)
Gun-involved violence	2.95	-29.2	12.7	(-54, -4)
Shooting victims	1.55	-35.0	11.2	(-57, -13)
Auto theft	1.03	-34.1	10.8	(-55, -13)
Property crime	28.66	-18.9	12.5	(-43, 6)
Other assault	18.82	-24.1	10.3	(-44, -4)
Panel B: Adjacent Tracts				
Violent crime	8.65	-9.0	10.9	(-30, 12)
Gun-involved violence	1.87	-10.6	11.6	(-33, 12)
Shooting victims	0.80	-22.3	9.9	(-42, -3)
Auto theft	1.68	-28.1	10.1	(-48, -8)
Property crime	46.30	-4.0	10.6	(-25, 17)
Other assault	13.41	-3.3	9.0	(-21, 14)

Notes: Panel A reports SDID estimates for the top 6 active tracts, excluding the target area. Panel B reports SDID estimates for adjacent border tracts (n=8). Panel A and B share 3 census tracts. Standard errors estimated using the placebo method.

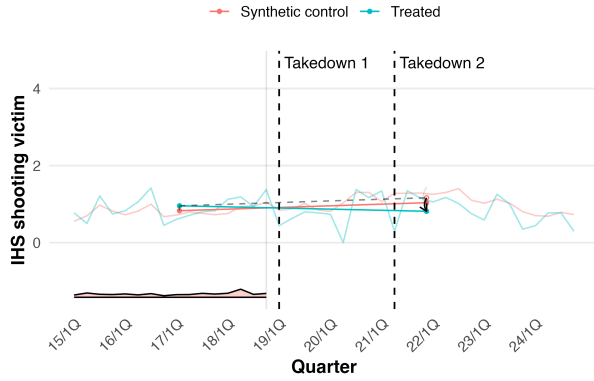
Appendix Tables and Figures



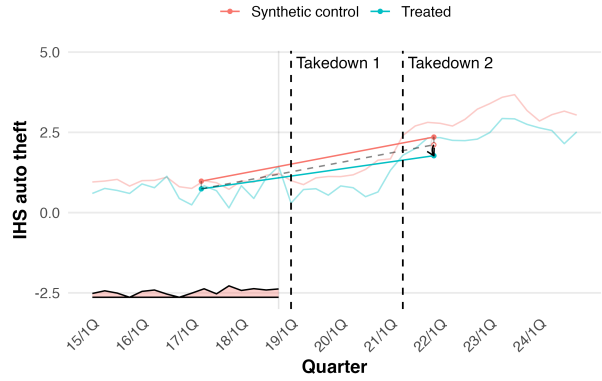
(a) Violent crime, RMSPE = 0.115



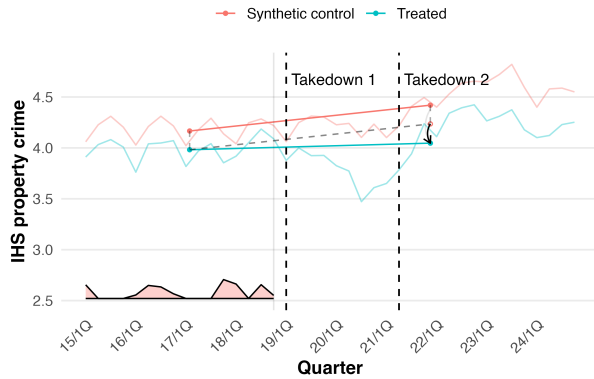
(b) Gun-involved violence, RMSPE = 0.267



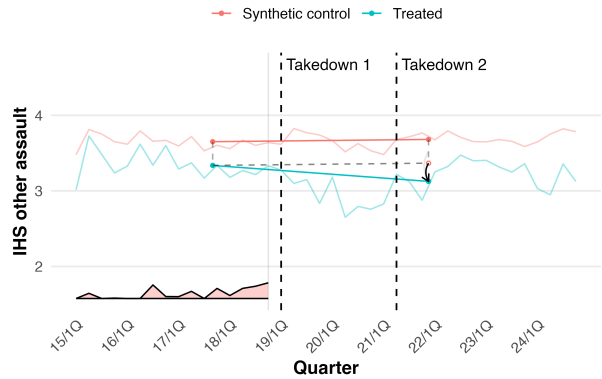
(c) Shooting victims, RMSPE = 0.198



(d) Auto theft, RMSPE = 0.201



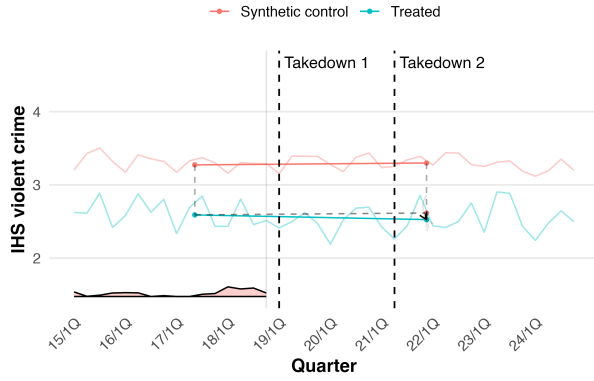
(e) Property crime, RMSPE = 0.054



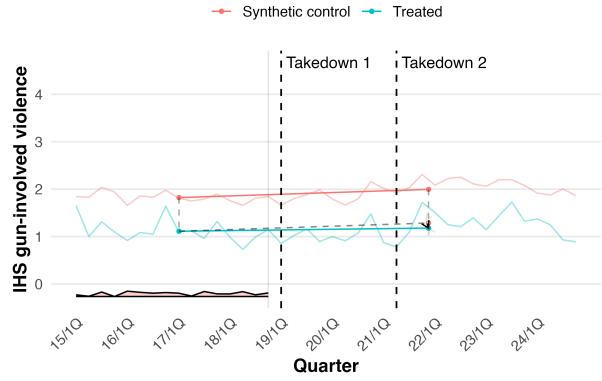
(f) Other assault, RMSPE = 0.108

Figure A1: Top 6 Active Tracts Synthetic Difference-in-Differences Plots

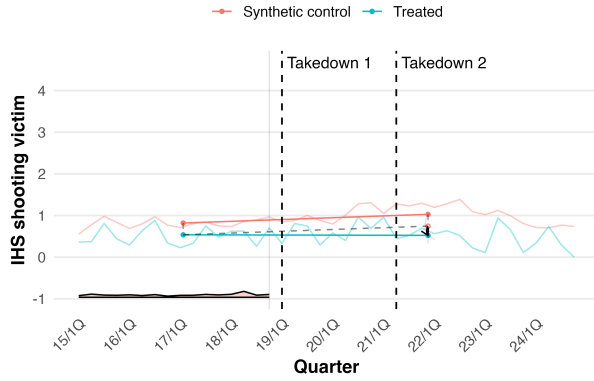
Note. RMSPE denotes the pre-treatment root mean squared prediction error. Lower values indicate better pre-treatment fit.



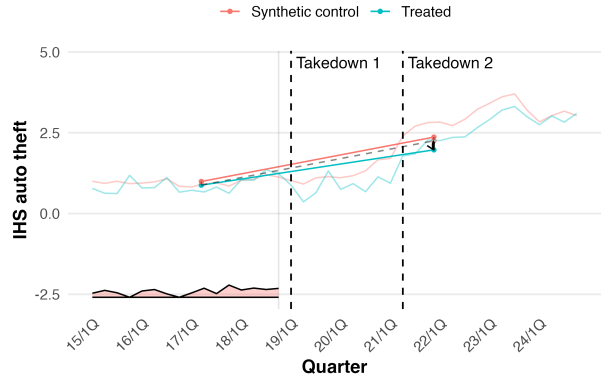
(a) Violent crime, RMSPE = 0.135



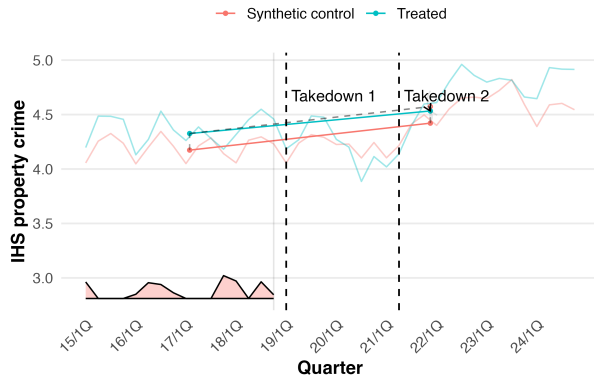
(b) Gun-involved violence, RMSPE = 0.185



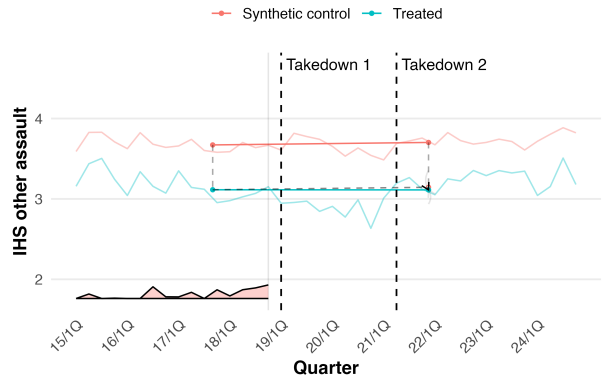
(c) Shooting victims, RMSPE = 0.159



(d) Auto theft, RMSPE = 0.170



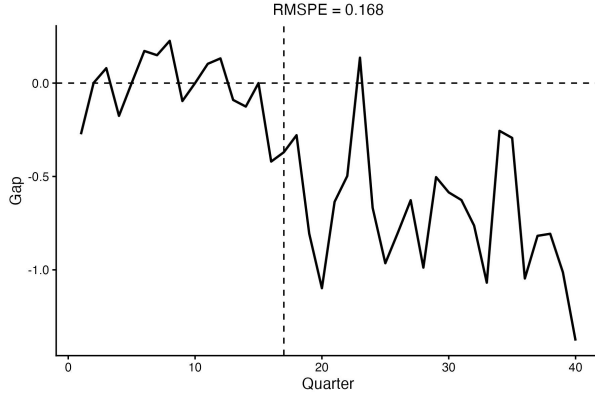
(e) Property crime, RMSPE = 0.076



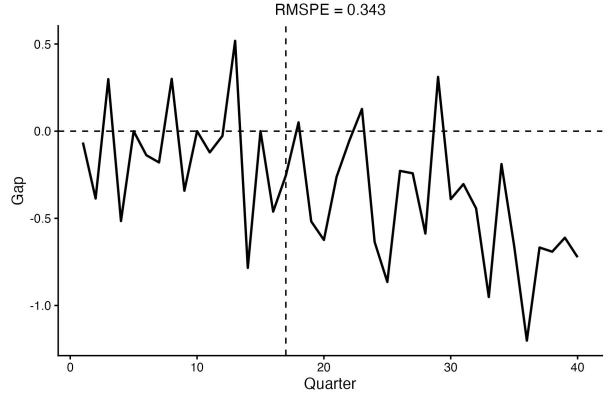
(f) Other assault, RMSPE = 0.103

Figure A2: Adjacent Tracts Synthetic Difference-in-Differences Plots

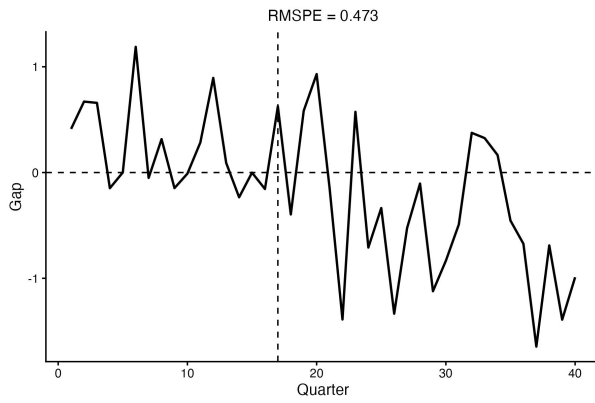
Note. RMSPE denotes the pre-treatment root mean squared prediction error. Lower values indicate better pre-treatment fit.



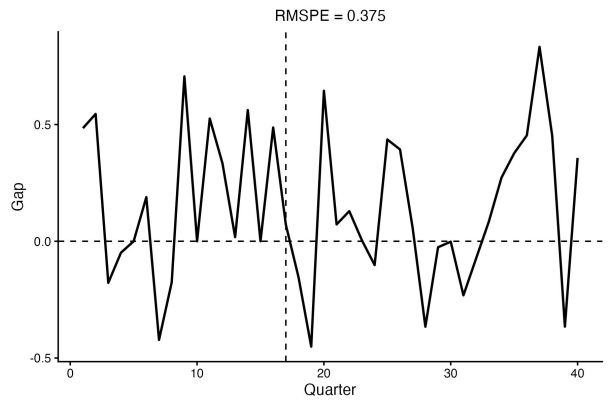
(a) Violent crime



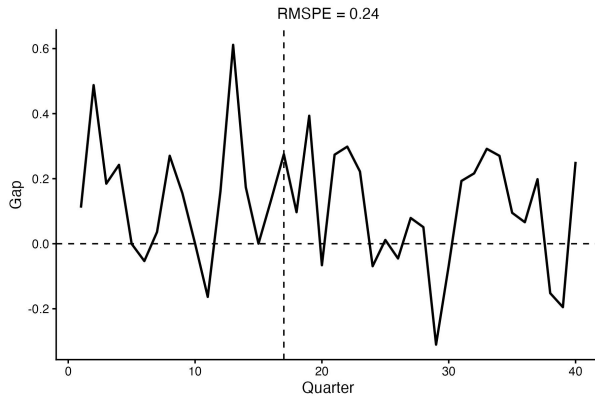
(b) Gun-involved violence



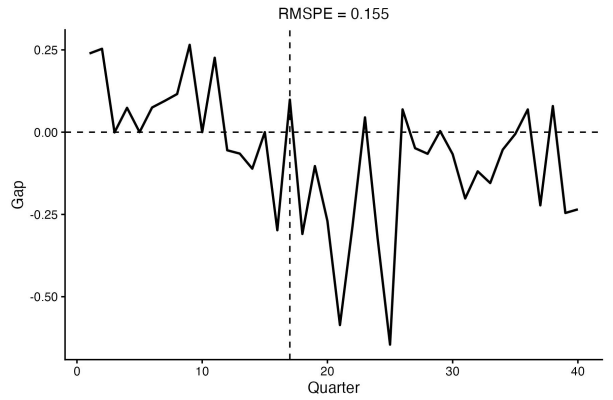
(c) Shooting victims



(d) Auto theft



(e) Property crime



(f) Other assault

Figure A3: Target Tracts Synthetic Control Method Gap Plots (Panel A in Table A7)

Note. RMSPE denotes the pre-treatment root mean squared prediction error. Lower values indicate better pre-treatment fit.

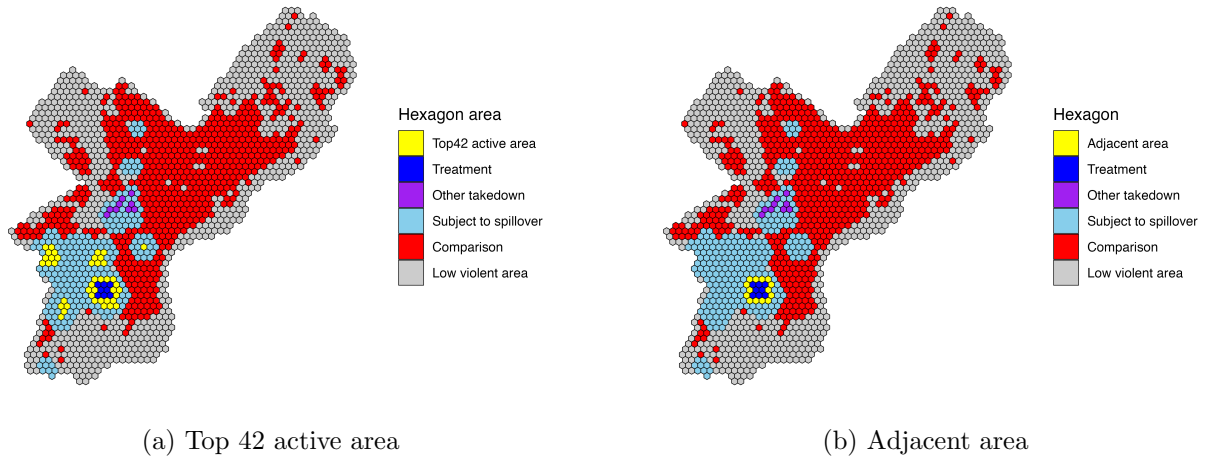


Figure A4: Hexagon Level Analysis Map

Table A1: SDID and SCM Estimates for Auto Theft in Airport Area

	Estimate (%)	SE (%)	95% CI (%)
Panel A: SDID			
Auto theft	-38	19.8	(-77, 1)
Panel B: SCM			
Pre mean	4.63		
Pre diff	3.45		
Takedown 1 DID (%)	-4.8		($p = 0.515$)
Takedown 2 DID (%)	-48.6		($p = 0.085$)
Total DID (%)	-35.1		($p = 0.108$)
N	130		

Notes: Panel A reports Synthetic Difference-in-Differences (SDID) estimates for auto theft in the airport area (n=2). Panel B reports Synthetic Control Method (SCM) estimates. Pre mean denotes the average of auto theft in two airport tracts. Standard errors estimated using the placebo method.

Table A2: Synthetic Difference-in-Differences Estimates: Target Area with Covariates

Outcome	Pre Mean	Neighborhood Covariates			Neighborhood + Crime Covariates		
		SDID (%)	SE (%)	95% CI (%)	SDID (%)	SE (%)	95% CI (%)
Violent crime	18.12	-48.2	14.1	(-76, -20)	-42.7	11.9	(-66, -19)
Gun-involved violence	4.33	-40.8	15.4	(-71, -11)	-35.3	14.3	(-63, -7)
Shooting victims	2.42	-28.6	12.7	(-53, -4)	-27.9	12.6	(-53, -3)
Auto theft	1.58	-15.1	13.3	(-41, 11)	-10.1	11.9	(-33, 14)
Property crime	33.78	-8.2	13.7	(-35, 19)	2.4	11.5	(-20, 25)
Other assault	24.67	-23.2	11.8	(-46, 0)	-14.3	10.2	(-34, 6)

Notes: Time-varying covariates are incorporated using a two-stage method. In the first stage, neighborhood-level and crime covariates are partialled out. The neighborhood variables includes population, poverty rate, white rate, median income, unemployment rate, public assistance household rate, foreign rate, and male rate in a given year. For crime covariates, the model includes the different types of crime in the first stage. The standard errors were adjusted by multiplying the estimated standard error from Synthetic DiD by $\frac{\sqrt{n-k}}{\sqrt{n-k+c}}$ where n is the number of observations, k is the number of variables in Synthetic DiD, and c is the number of time-varying covariates included in the first stage.

Table A3: Synthetic Difference-in-Differences Estimates: Spillover Effects with Covariates

Outcome	Pre Mean	Neighborhood Covariates			Neighborhood + Crime Covariates		
		SDID (%)	SE (%)	95% CI (%)	SDID (%)	SE (%)	95% CI (%)
Panel A: Top 6 Active Tracts							
Violent crime	13.85	-27.5	11.6	(-50, -5)	-20.7	10.0	(-40, -1)
Gun-involved violence	2.95	-26.2	12.4	(-50, -2)	-19.9	11.5	(-42, 3)
Shooting victims	1.55	-30.9	10.9	(-52, -10)	-29.9	10.8	(-51, -9)
Auto theft	1.03	-31.7	10.8	(-53, -11)	-28.3	10.1	(-48, -9)
Property crime	28.66	-17.3	11.1	(-39, 5)	-8.2	9.5	(-27, 10)
Other assault	18.82	-27.0	9.4	(-45, -9)	-17.4	7.9	(-33, -2)
Panel B: Border Tracts							
Violent crime	8.65	-8.4	10.9	(-30, 13)	-9.7	9.3	(-28, 9)
Gun-involved violence	1.87	-9.3	11.0	(-31, 12)	-9.6	10.4	(-30, 11)
Shooting victims	0.80	-20.7	9.9	(-40, -1)	-19.1	9.8	(-38, 0)
Auto theft	1.68	-30.8	9.4	(-49, -12)	-30.3	9.3	(-48, -12)
Property crime	46.30	-4.0	11.1	(-26, 18)	-1.8	9.5	(-20, 17)
Other assault	13.41	-5.4	9.0	(-23, 12)	-6.1	7.6	(-21, 9)

Notes: Time-varying covariates are incorporated using a two-stage method. In the first stage, neighborhood-level and crime covariates are partialled out. The neighborhood variables includes population, poverty rate, white rate, median income, unemployment rate, public assistance household rate, foreign rate, and male rate. For crime covariates, the model includes the different types of crime only in the first stage. The standard errors were adjusted by multiplying the estimated standard error from Synthetic DiD by $\frac{\sqrt{n-k}}{\sqrt{n-k+c}}$ where n is the number of observations, k is the number of variables in Synthetic DiD, and c is the number of time-varying covariates included in the first stage.

Table A4: Synthetic Control Method: Top 6 Active Tracts

Sample	Pre mean	Pre diff	Takedwon 1			Takedwon 2			Total			N
			Post diff	DID (%)	<i>p</i>	Post diff	DID (%)	<i>p</i>	Post diff	DID (%)	<i>p</i>	
Panel A: Outcome lags												
Violent crime	13.85	-8.3	-63.5	-55.3	0.075	-59.9	-51.6	0.050	-53.9	-45.6	0.050	80
Gun-involved violence	2.95	-8.7	-55.5	-46.7	0.193	-56.9	-48.1	0.083	-52.4	-43.7	0.083	109
Shooting victims	1.55	13.4	-23.2	-36.7	0.267	-27.0	-40.5	0.333	-19.6	-33.1	0.267	15
Auto theft	1.03	-19.9	-53.8	-33.9	0.253	-61.9	-42.0	0.115	-54.8	-34.9	0.126	87
Property crime	28.66	-1.4	-33.0	-31.6	0.250	-21.8	-20.4	0.250	-25.7	-24.2	0.250	4
Other assault	18.82	1.7	-63.0	-64.7	0.020	-37.1	-38.8	0.059	-47.1	-48.8	0.020	51
Panel B: Outcome lags + Neighborhood Covariates												
Violent crime	13.85	-11.2	-45.4	-34.2	0.192	-51.6	-40.4	0.080	-45.7	-34.5	0.104	125
Gun-involved violence	2.95	-21.5	-62.7	-41.2	0.150	-52.0	-30.5	0.203	-56.2	-34.7	0.120	133
Shooting victims	1.55	5.9	-33.3	-39.2	0.161	-42.1	-47.9	0.194	-33.3	-39.2	0.097	31
Auto theft	1.03	-13.1	-45.6	-32.5	0.299	-69.6	-56.4	0.052	-57.9	-44.7	0.113	97
Property crime	28.66	-1.1	-27.5	-26.4	0.222	-16.1	-15.0	0.111	-21.9	-20.7	0.111	9
Other assault	18.82	0.1	-77.3	-77.3	0.012	-47.9	-48.0	0.012	-56.4	-56.5	0.012	84
Panel C: Outcome lags + Neighborhood Covariates + Other Crime types												
Violent crime	13.85	-0.9	-59.9	-59.0	0.040	-58.1	-57.2	0.020	-51.8	-50.9	0.020	99
Gun-involved violence	2.95	-6.0	-116.2	-110.2	0.035	-81.1	-75.1	0.009	-77.3	-71.3	0.009	114
Shooting victims	1.55	0.0	-36.7	-36.7	0.164	-49.1	-49.1	0.109	-37.4	-37.4	0.073	55
Auto theft	1.03	-1.4	-43.5	-42.1	0.273	-61.4	-60.0	0.091	-51.1	-49.6	0.136	22
Property crime	28.66	-0.3	-23.5	-23.2	0.500	-18.8	-18.5	0.500	-23.8	-23.5	0.500	2
Other assault	18.82	-1.6	-65.5	-63.9	0.010	-35.7	-34.1	0.071	-49.3	-47.7	0.010	98

Notes: The SCM estimates are computed for the top 6 active tracts using predictors including lagged outcomes at 2016/1Q, 2017/2Q, and 2018/3Q. Panels B and C additionally incorporate neighborhood covariates and other crime types, respectively. The SCM analysis restricts donor units to those with pre-treatment RMSPE less than or equal to two times that of the treated unit. The DID estimates for Takedwon 1 and Takedwon 2 capture the two-year periods following each intervention. The Total column reports the estimate using the full six-year post-intervention period following Takedwon 1 (2019 Q1).

Table A5: Synthetic Control Method: Adjacent Area

Sample	Pre mean	Pre diff	Takedown 1			Takedown 2			Total			N
			Post diff	DID (%)	p	Post diff	DID (%)	p	Post diff	DID (%)	p	
Panel A: Outcome lags												
Violent crime	8.65	-5.3	-76.7	-71.4	0.028	-50.5	-45.2	0.093	-46.9	-41.7	0.056	107
Gun-involved violence	1.87	0.8	-109.9	-110.6	0.062	-57.7	-58.4	0.062	-62.7	-63.4	0.062	16
Shooting victims	0.80	9.7	-1.3	-11.1	1.000	-32.1	-41.9	1.000	-12.8	-22.6	1.000	1
Auto theft	1.68	-4.0	-31.9	-27.9	0.500	-52.9	-48.9	0.250	-36.6	-32.6	0.500	4
Property crime	46.30	0.0	-14.7	-14.7	0.432	-0.9	-1.0	0.591	-4.2	-4.2	0.614	44
Other assault	13.41	8.3	-79.3	-87.6	0.016	-34.4	-42.7	0.081	-42.3	-50.6	0.041	123
Panel B: Outcome lags + Neighborhood Covariates												
Violent crime	8.65	-2.5	-60.7	-58.2	0.054	-64.6	-62.1	0.009	-40.5	-38.0	0.045	111
Gun-involved violence	1.87	-4.0	-64.5	-60.5	0.075	-69.6	-65.6	0.030	-54.3	-50.3	0.060	67
Shooting victims	0.80	9.7	6.8	-2.9	0.460	-29.7	-39.4	0.180	-8.3	-18.0	0.280	50
Auto theft	1.68	-0.9	-15.6	-14.7	0.667	-44.1	-43.2	0.333	-25.6	-24.7	0.500	6
Property crime	46.30	1.0	-31.6	-32.6	0.111	-15.3	-16.3	0.204	-19.4	-20.4	0.148	54
Other assault	13.41	0.7	-58.7	-59.5	0.023	-14.3	-15.1	0.310	-24.8	-25.5	0.103	87
Panel C: Outcome lags + Neighborhood Covariates + Other Crime types												
Violent crime	8.65	-0.4	-54.7	-54.3	0.053	-52.1	-51.7	0.039	-37.5	-37.2	0.039	76
Gun-involved violence	1.87	-3.8	-68.9	-65.1	0.024	-63.0	-59.2	0.048	-53.5	-49.8	0.024	42
Shooting victims	0.80	3.4	-0.3	-3.7	0.500	-30.2	-33.6	0.188	-12.4	-15.8	0.312	16
Auto theft	1.68	0.1	-24.3	-24.4	1.000	-40.2	-40.3	1.000	-30.7	-30.8	1.000	1
Property crime	46.30	0.2	-21.9	-22.1	0.282	1.1	1.0	0.692	-2.3	-2.5	0.577	78
Other assault	13.41	-1.7	-71.4	-69.7	0.010	-26.9	-25.3	0.117	-35.7	-34.0	0.029	103

Notes: The SCM estimates are computed for the adjacent tracts using predictors including lagged outcomes at 2016/1Q, 2017/2Q, and 2018/3Q. Panels B and C additionally incorporate neighborhood covariates and other crime types, respectively. The SCM analysis restricts donor units to those with pre-treatment RMSPE less than or equal to two times that of the treated unit. The DID estimates for Takedown 1 and Takedown 2 capture the two-year periods following each intervention. The Total column reports the estimate using the full six-year post-intervention period following Takedown 1 (2019 Q1).

Table A6: Traditional Difference-in-Differences Estimates for Target Area

Sample	Pre Mean	Takedown 1				Takedown 2				Total			
		DID (%)	SE (%)	95% CI (%)	<i>p</i>	DID (%)	SE (%)	95% CI (%)	<i>p</i>	DID (%)	SE (%)	95% CI (%)	<i>p</i>
Panel A: No Control variables													
Violent crime	18.12	-34.1	13.8	(-61.1, -7.1)	0.013	-47.2	13.4	(-73.5, -20.9)	0.000	-46.2	10.2	(-66.2, -26.2)	0.000
Gun-involved violence	4.33	-28.8	18.2	(-64.5, 6.9)	0.113	-34.4	17.4	(-68.5, -0.3)	0.048	-39.3	13.1	(-65, -13.6)	0.003
Shooting victims	2.42	1.1	19.5	(-37.1, 39.3)	0.956	-48.6	19.7	(-87.2, -10)	0.014	-29.0	14.8	(-58, 0)	0.049
Auto theft	1.58	-12.0	16.6	(-44.5, 20.5)	0.471	-25.7	15.7	(-56.5, 5.1)	0.102	-8.4	16.3	(-40.3, 23.5)	0.608
Property crime	33.78	2.3	12.8	(-22.8, 27.4)	0.856	-4.6	12.8	(-29.7, 20.5)	0.716	-3.0	9.8	(-22.2, 16.2)	0.761
Other assault	24.67	-32.6	11.7	(-55.5, -9.7)	0.006	-21.0	11.6	(-43.7, 1.7)	0.070	-27.2	8.6	(-44.1, -10.3)	0.001
Panel B: Neighborhood control variables													
Violent crime	18.12	-15.7	11.9	(-39, 7.6)	0.186	-23.8	11.7	(-46.7, -0.9)	0.042	-26.9	9.1	(-44.7, -9.1)	0.003
Gun-involved violence	4.33	-9.4	15.8	(-40.4, 21.6)	0.550	-7.8	15.4	(-38, 22.4)	0.614	-16.4	11.8	(-39.5, 6.7)	0.164
Shooting victims	2.42	26.6	16.7	(-6.1, 59.3)	0.111	-12.4	17.0	(-45.7, 20.9)	0.463	3.1	12.9	(-22.2, 28.4)	0.809
Auto theft	1.58	-7.5	16.0	(-38.9, 23.9)	0.641	-19.8	15.0	(-49.2, 9.6)	0.186	-2.4	16.0	(-33.8, 29)	0.881
Property crime	33.78	6.2	11.7	(-16.7, 29.1)	0.600	-7.4	11.7	(-30.3, 15.5)	0.528	-8.6	9.0	(-26.2, 9)	0.339
Other assault	24.67	-19.0	10.8	(-40.2, 2.2)	0.079	-5.6	10.7	(-26.6, 15.4)	0.599	-13.8	7.9	(-29.3, 1.7)	0.083

Notes: Standard errors are clustered at census tracts. Panel A does not include control variables. Panel B uses neighborhood-level control variables, including population, White rate, unemployment rate, male, income, poverty rate, foreign-born rate, and public assistance rate of each tract in a given year. The DID estimates for Takedown 1 and Takedown 2 capture the two-year periods following each intervention. The Total column reports the estimate using the full six-year post-intervention period following Takedown 1 (2019 Q1).

Table A7: Synthetic Control Method for Target Tracts

Sample	Pre mean	Pre diff	Takedown 1			Takedown 2			Total			N
			Post diff	DID (%)	<i>p</i>	Post diff	DID (%)	<i>p</i>	Post diff	DID (%)	<i>p</i>	
Panel A: Outcome lags												
Violent crime	18.12	-2.0	-52.7	-50.7	0.077	-74.5	-72.5	0.011	-69.8	-67.8	0.011	91
Gun-involved violence	4.33	-11.9	-27.1	-15.2	0.450	-35.4	-23.5	0.230	-44.2	-32.3	0.140	100
Shooting victims	2.42	23.6	0.9	-22.6	0.365	-46.4	-69.9	0.083	-40.2	-63.8	0.031	96
Auto theft	1.58	18.9	2.6	-16.3	0.456	-2.1	-21.0	0.263	11.9	-7.0	0.447	114
Property crime	33.78	14.7	17.8	3.1	0.780	5.1	-9.6	0.553	9.9	-4.8	0.636	132
Other assault	24.67	5.1	-21.6	-26.7	0.180	-7.3	-12.4	0.315	-14.9	-20.0	0.124	89
Panel B: Outcome lags + Neighborhood Covariates												
Violent crime	18.12	-5.5	-31.6	-26.0	0.220	-39.6	-34.0	0.154	-43.5	-38.0	0.057	123
Gun-involved violence	4.33	-22.6	-44.4	-21.8	0.341	-38.5	-15.9	0.364	-51.2	-28.6	0.152	132
Shooting victims	2.42	3.1	-8.5	-11.7	0.358	-55.8	-58.9	0.075	-38.4	-41.5	0.075	67
Auto theft	1.58	5.6	-6.6	-12.2	0.415	-29.2	-34.8	0.146	-8.2	-13.8	0.323	130
Property crime	33.78	2.9	7.4	4.6	0.697	4.0	1.1	0.615	4.4	1.5	0.648	122
Other assault	24.67	1.4	-37.4	-38.8	0.079	-24.0	-25.3	0.134	-32.0	-33.4	0.055	127
Panel C: Outcome lags + Neighborhood Covariates + Other Crime types												
Violent crime	18.12	-0.3	-49.8	-49.6	0.058	-51.6	-51.3	0.033	-52.6	-52.3	0.017	120
Gun-involved violence	4.33	-14.3	-58.9	-44.6	0.174	-41.0	-26.7	0.242	-55.7	-41.5	0.083	132
Shooting victims	2.42	12.0	0.0	-12.1	0.380	-38.8	-50.9	0.130	-27.1	-39.2	0.080	100
Auto theft	1.58	1.8	-16.2	-18.0	0.565	-13.7	-15.6	0.522	-1.5	-3.3	0.609	23
Property crime	33.78	0.4	1.7	1.3	0.706	-2.3	-2.7	0.548	-0.1	-0.5	0.587	126
Other assault	24.67	1.4	-40.1	-41.5	0.027	-23.4	-24.8	0.134	-33.4	-34.8	0.045	112

Notes: The SCM estimates are computed for the target tracts using predictors including lagged outcomes at 2016/1Q, 2017/2Q, and 2018/3Q. Panels B and C additionally incorporate neighborhood covariates and other crime types, respectively. The SCM analysis restricts donor units to those with pre-treatment RMSPE less than or equal to two times that of the treated unit. The DID estimates for Takedown 1 and Takedown 2 capture the two-year periods following each intervention. The Total column reports the estimate using the full six-year post-intervention period following Takedown 1 (2019 Q1).

Table A8: SCM Inference under Different RMSPE Thresholds

Outcome	3 × RMSPE Threshold				5 × RMSPE Threshold			
	p (Takedown 1)	p (Takedown 2)	p (Total)	N	p (Takedown 1)	p (Takedown 2)	p (Total)	N
Panel A: Outcome lag								
Violent crime	0.103	0.008	0.016	126	0.098	0.008	0.015	133
Gun-involved violence	0.436	0.226	0.135	133	0.433	0.231	0.134	134
Shooting victims	0.340	0.085	0.038	106	0.340	0.085	0.038	106
Auto theft	0.427	0.244	0.435	131	0.427	0.244	0.435	131
Property crime	0.782	0.549	0.632	133	0.782	0.549	0.632	133
Other assault	0.183	0.351	0.176	131	0.195	0.353	0.180	133
Panel B: Outcome lags + Neighborhood Covariates								
Violent crime	0.229	0.160	0.061	131	0.241	0.158	0.060	133
Gun-involved violence	0.338	0.368	0.150	133	0.338	0.368	0.150	133
Shooting victims	0.346	0.055	0.079	127	0.358	0.060	0.082	134
Auto theft	0.403	0.142	0.313	134	0.403	0.142	0.313	134
Property crime	0.712	0.621	0.667	132	0.714	0.617	0.662	133
Other assault	0.083	0.135	0.060	133	0.083	0.135	0.060	133
Panel C: Outcome lags + Neighborhood Covariates + Other Crime types								
Violent crime	0.069	0.046	0.023	131	0.075	0.045	0.023	133
Gun-involved violence	0.172	0.246	0.082	134	0.172	0.246	0.082	134
Shooting victims	0.388	0.127	0.090	134	0.388	0.127	0.090	134
Auto theft	0.388	0.345	0.483	116	0.373	0.321	0.478	134
Property crime	0.710	0.550	0.595	131	0.707	0.549	0.594	133
Other assault	0.045	0.143	0.060	133	0.045	0.143	0.060	133

Notes: This table reports placebo-based p -values and the number of valid donor units (N) for the Synthetic Control Method (SCM) under two RMSPE thresholds. The left panel restricts donor units to those with pre-treatment RMSPE less than or equal to three times that of the treated unit, while the right panel uses a five-times threshold. Panels correspond to different model specifications. Takedown 1 and Takedown 2 represent post-intervention periods, and Total reflects the full post-treatment period.

Table A9: Synthetic Difference-in-Differences Estimates for Target Area with Different Comparison Groups

Outcome	Pre mean	SDID (%)	SE (%)	95% CI (%)
Panel A: Top 25% Violent Crime Comparison Tracts				
Violent crime	18.12	-44.4	8.1	(-60, -28)
Gun-involved violence	4.33	-25.6	13.7	(-52, 1)
Shooting victims	2.42	-36.7	16.5	(-69, -4)
Auto theft	1.58	-20.4	16.2	(-52, 11)
Property crime	33.78	-6.9	10.8	(-28, 14)
Other assault	24.67	-10.0	8.1	(-26, 6)
Panel B: Top 75% Violent Crime Comparison Tracts				
Violent crime	18.12	-47.2	13.9	(-74, -20)
Gun-involved violence	4.33	-39.4	14.5	(-68, -11)
Shooting victims	2.42	-30.1	12.6	(-55, -6)
Auto theft	1.58	-6.4	14.1	(-34, 21)
Property crime	33.78	-9.7	12.8	(-35, 15)
Other assault	24.67	-17.8	11.9	(-41, 5)
Panel C: All Comparison Tracts				
Violent crime	18.12	-45.6	13.6	(-72, -19)
Gun-involved violence	4.33	-37.5	12.6	(-62, -13)
Shooting victims	2.42	-28.0	10.9	(-49, -7)
Auto theft	1.58	1.4	16.6	(-31, 34)
Property crime	33.78	-5.7	12.4	(-30, 19)
Other assault	24.67	-23.4	12.8	(-49, 2)

Notes: Panel A uses the top 25% of census tracts as the comparison group (n=60). Panel B uses the top 75% of census tracts as the comparison group (n=209). Panel C uses all census tracts as the comparison group (n=299). All models here exclude possible spillover areas as our main specification.

Table A10: Synthetic Difference-in-Differences Estimates for Target Area under Different Thresholds for Gang Active Areas

Outcome	Pre mean	SDID (%)	SE (%)	95% CI (%)
Panel A: Excluding Top 10 Gang Areas + Adjacent Tracts				
Violent crime	18.12	-46.5	13.1	(-72, -21)
Gun-involved violence	4.33	-40.2	15.6	(-71, -10)
Shooting victims	2.42	-31.0	14.2	(-59, -3)
Auto theft	1.58	-12.5	14.7	(-41, 16)
Property crime	33.78	-2.6	13.4	(-29, 24)
Other assault	24.67	-17.9	12.9	(-43, 7)
Panel B: Excluding Top 50 Gang Areas + Adjacent Tracts				
Violent crime	18.12	-48.2	18.1	(-84, -13)
Gun-involved violence	4.33	-45.0	17.0	(-78, -12)
Shooting victims	2.42	-30.5	14.7	(-59, -2)
Auto theft	1.58	-13.1	13.2	(-39, 13)
Property crime	33.78	-11.4	15.8	(-42, 19)
Other assault	24.67	-13.9	12.4	(-38, 10)
Panel C: Excluding the Kensington area				
Violent crime	18.12	-47.8	17.1	(-81, -14)
Gun-involved violence	4.33	-40.5	15.0	(-70, -11)
Shooting victims	2.42	-35.1	14.9	(-64, -6)
Auto theft	1.58	-14.9	13.8	(-42, 12)
Property crime	33.78	-9.0	16.3	(-41, 23)
Other assault	24.67	-13.2	14.9	(-42, 16)

Notes: Panel A excludes the top 10 gang-active census tracts and their adjacent tracts from the comparison group, yielding 163 tracts for comparison group. Panel B excludes the top 50 gang-active census tracts and their adjacent tracts, yielding 106 census tracts for comparison group. Panel C excludes the census tracts that are identified as Kensington area from the primary model, resulting in 114 census tracts as comparison group.

Table A11: Synthetic Difference-in-Differences Estimates from Hexagon Level Analysis

Outcome	Pre Mean	SDID (%)	SE (%)	95% CI (%)
Panel A: Target Area (n= 8)				
Violent crime	5.50	-32.5	10.9	(-54, -11)
Gun-involved violence	1.27	-19.0	8.5	(-36, -2)
Shooting victims	0.84	-23.2	5.5	(-34, -12)
Auto theft	0.52	4.5	12.4	(-20, 29)
Property crime	10.46	-7.5	11.9	(-31, 16)
Other assault	7.50	-9.4	11.7	(-32, 14)
Panel B: Top 42 Active area (n=42)				
Violent crime	4.95	-14.3	4.6	(-23, -5)
Gun-involved violence	1.16	-9.2	3.9	(-17, -2)
Shooting victims	0.43	-0.8	2.2	(-5, 4)
Auto theft	0.43	-1.0	5.2	(-11, 9)
Property crime	12.33	-4.7	5.3	(-15, 6)
Other assault	7.03	-9.6	4.6	(-19, -1)
Panel C: Adjacent Area (n=14)				
Violent crime	3.38	-27.0	8.6	(-44, -10)
Gun-involved violence	0.74	-16.8	6.6	(-30, -4)
Shooting victims	0.39	-10.2	4.2	(-18, -2)
Auto theft	0.31	-29.6	9.6	(-48, -11)
Property crime	7.28	-5.3	9.7	(-24, 14)
Other assault	4.55	-29.7	7.9	(-45, -14)

Notes: The unit of analysis is a hexagon with a width of 500m in Philadelphia. Panel A reports estimates for the target area. Panel B reports estimates for the top 42 most active hexagons. Panel C reports estimates for hexagons adjacent to the target area. 703 hexagons are used as a comparison group.